

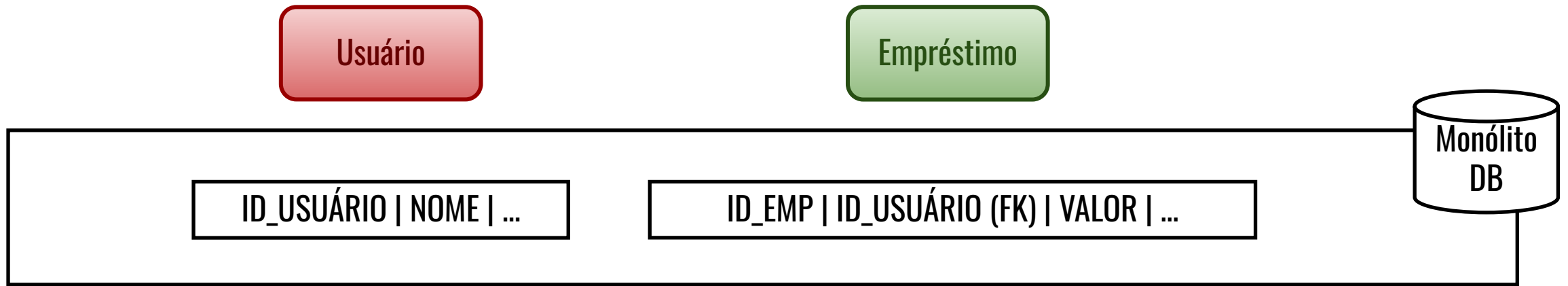
Arquitetura de Microserviços

Data Management

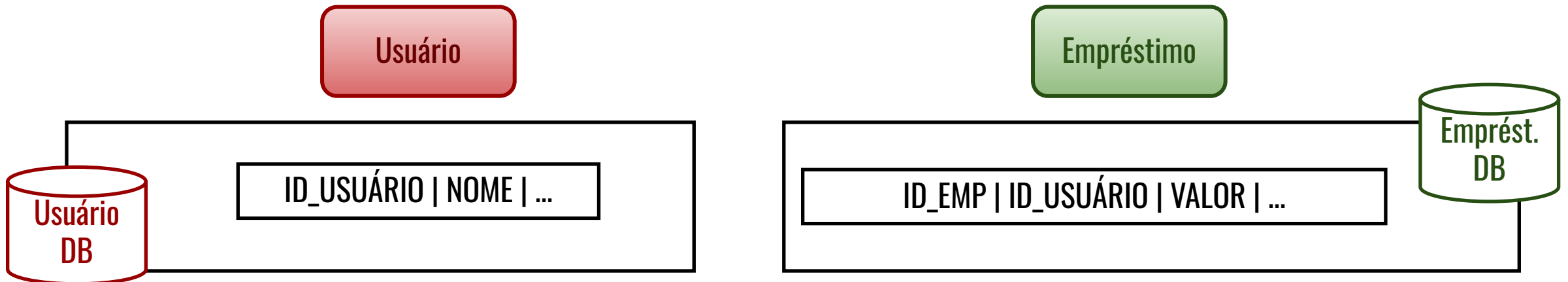
Prof. Luiz Gustavo Roos
luizroos@gmail.com

Tabelas Compartilhadas

Cenário Monólito

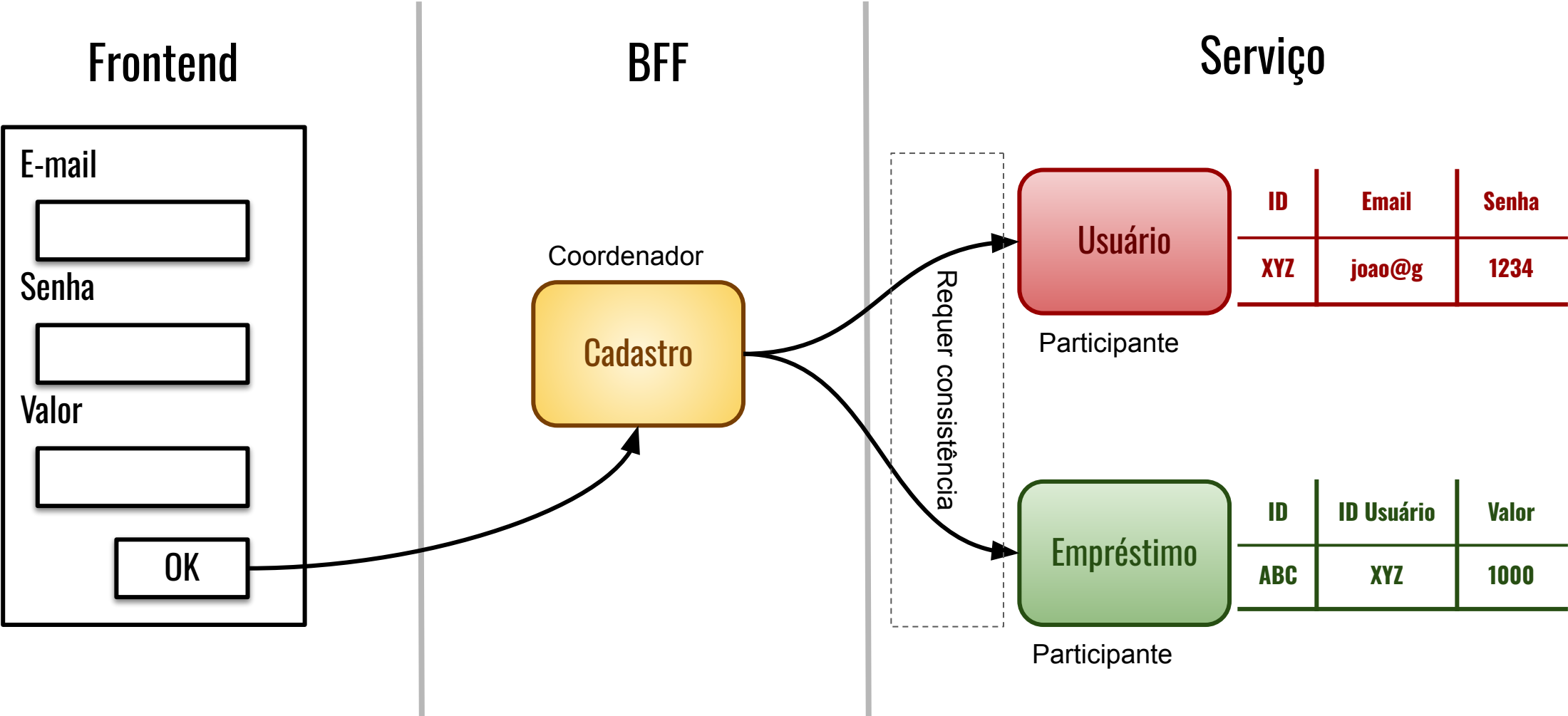


Cenário Microservices



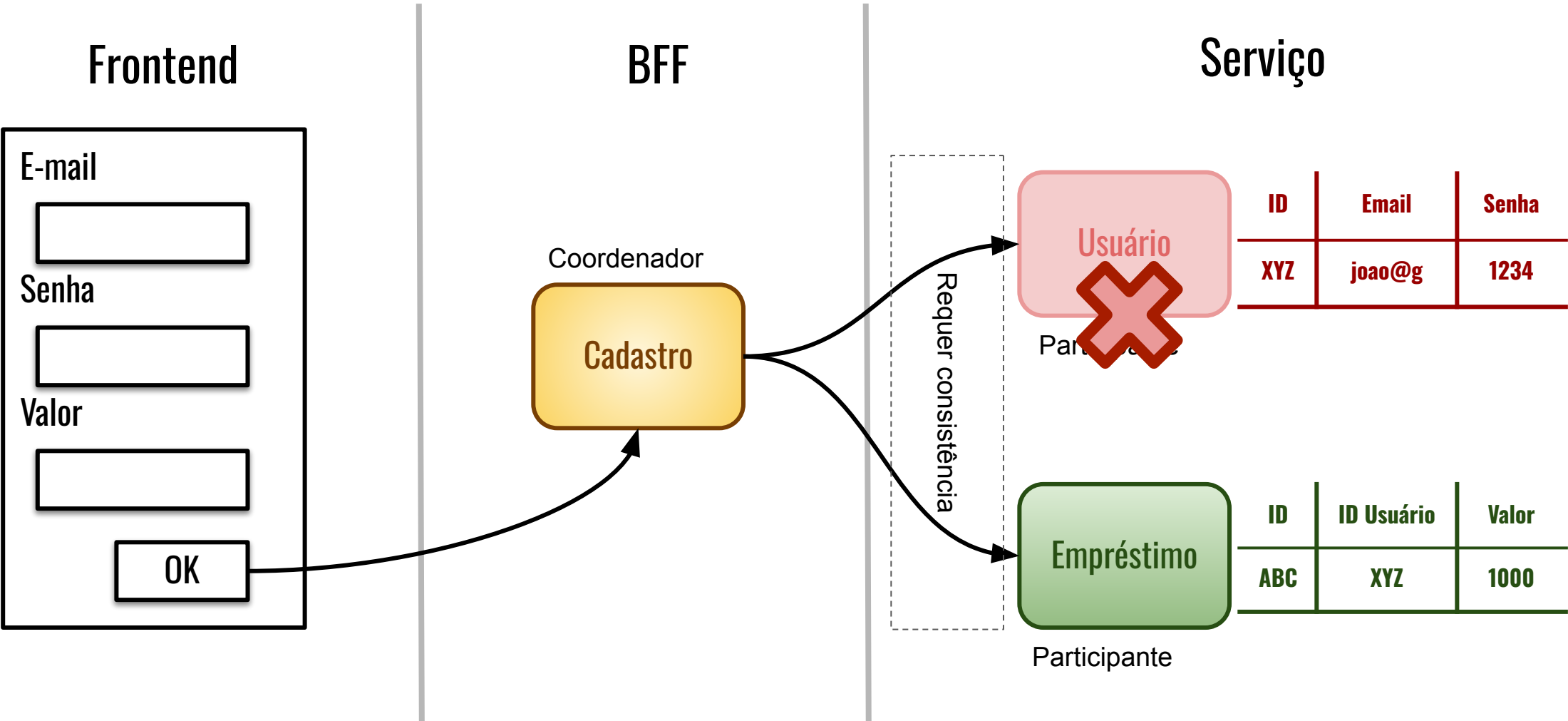
Transações distribuídas

Protocolo 2PC



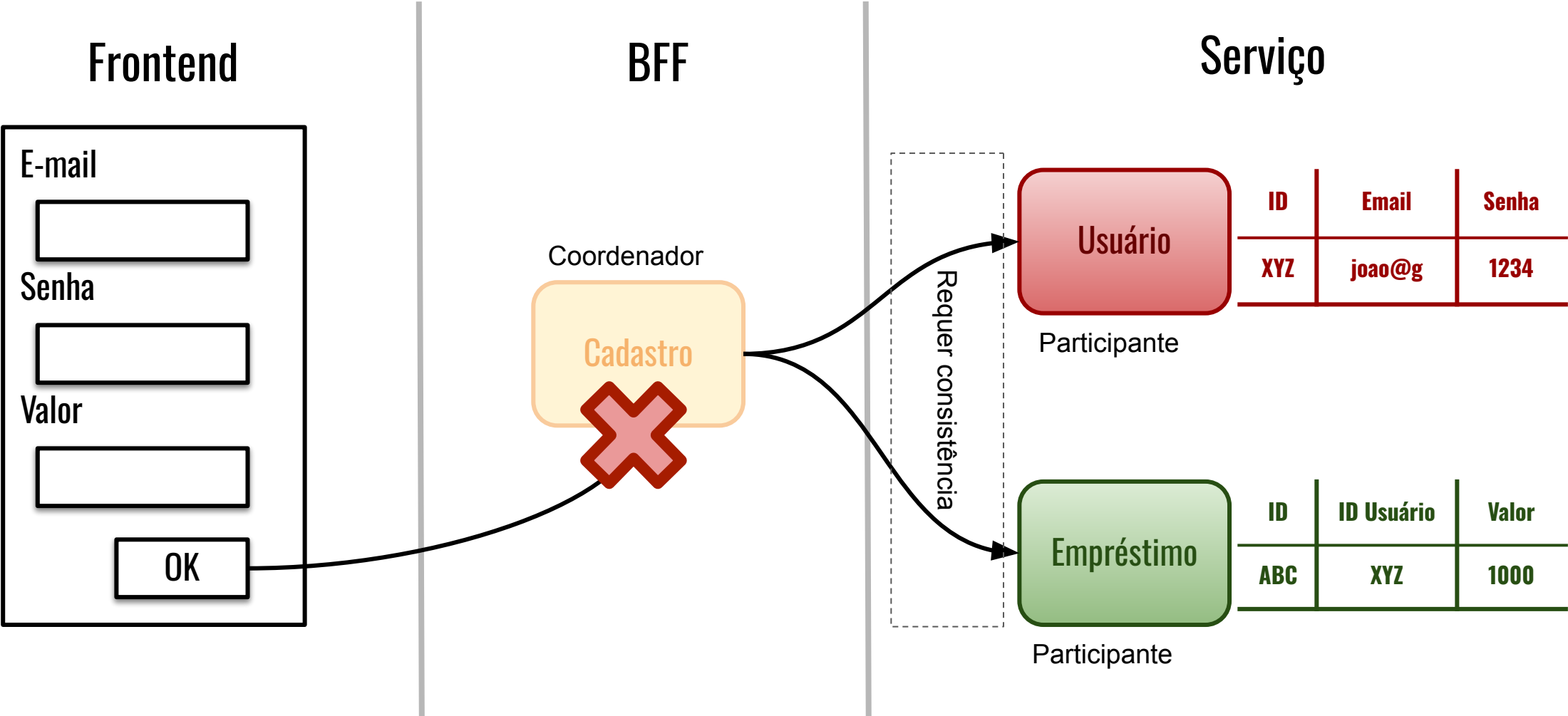
Transações distribuídas

Protocolo 2PC

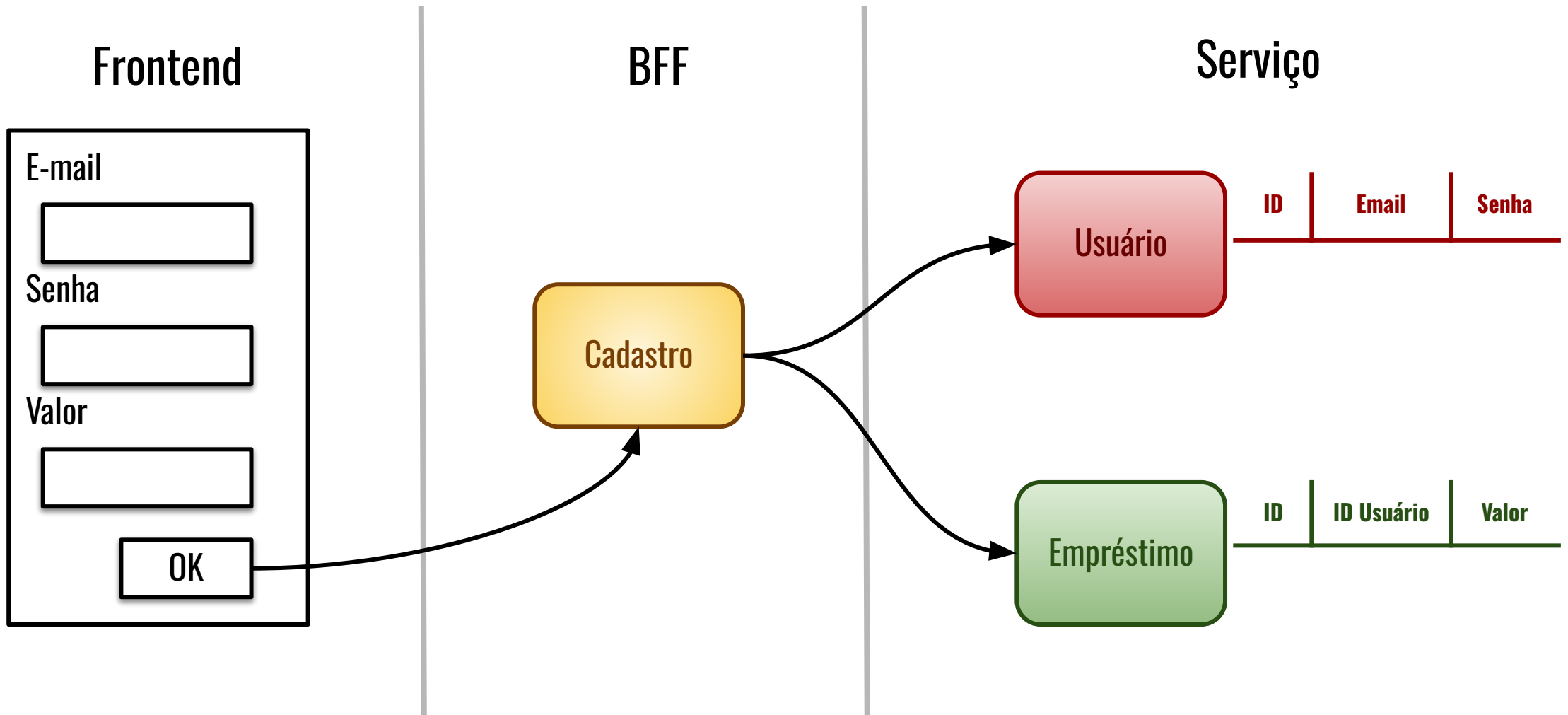


Transações distribuídas

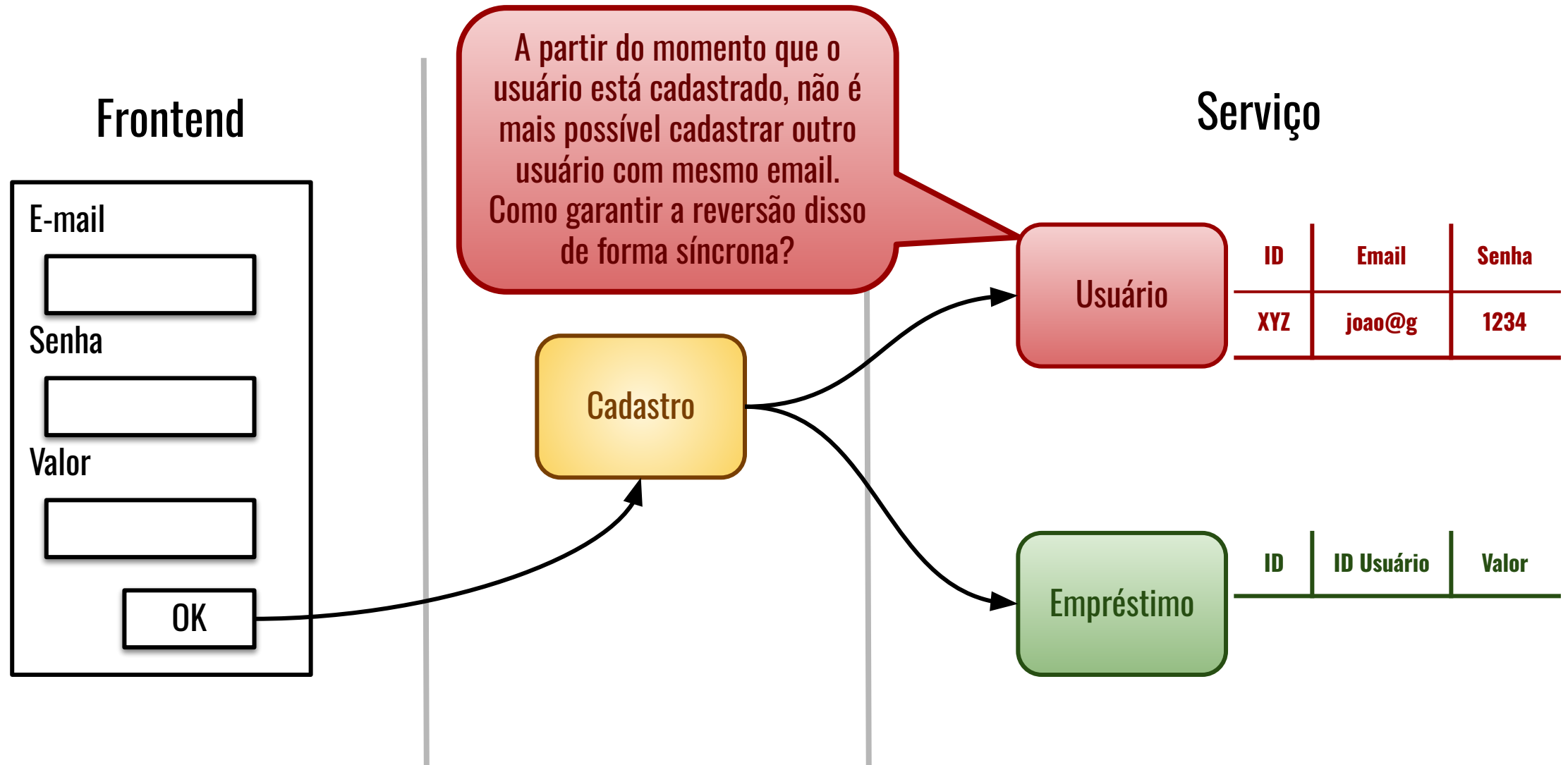
Protocolo 2PC



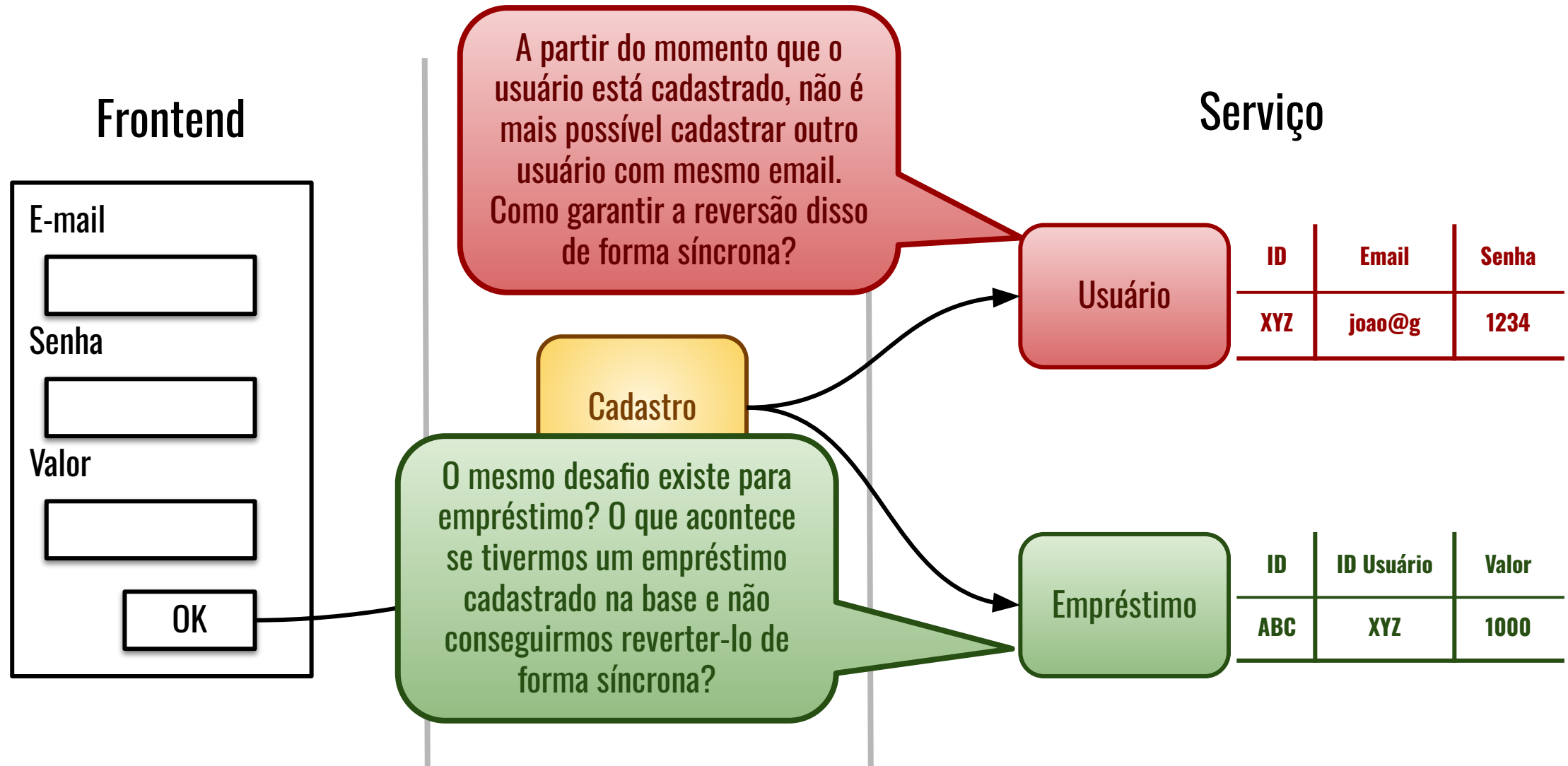
Modelando sem transação



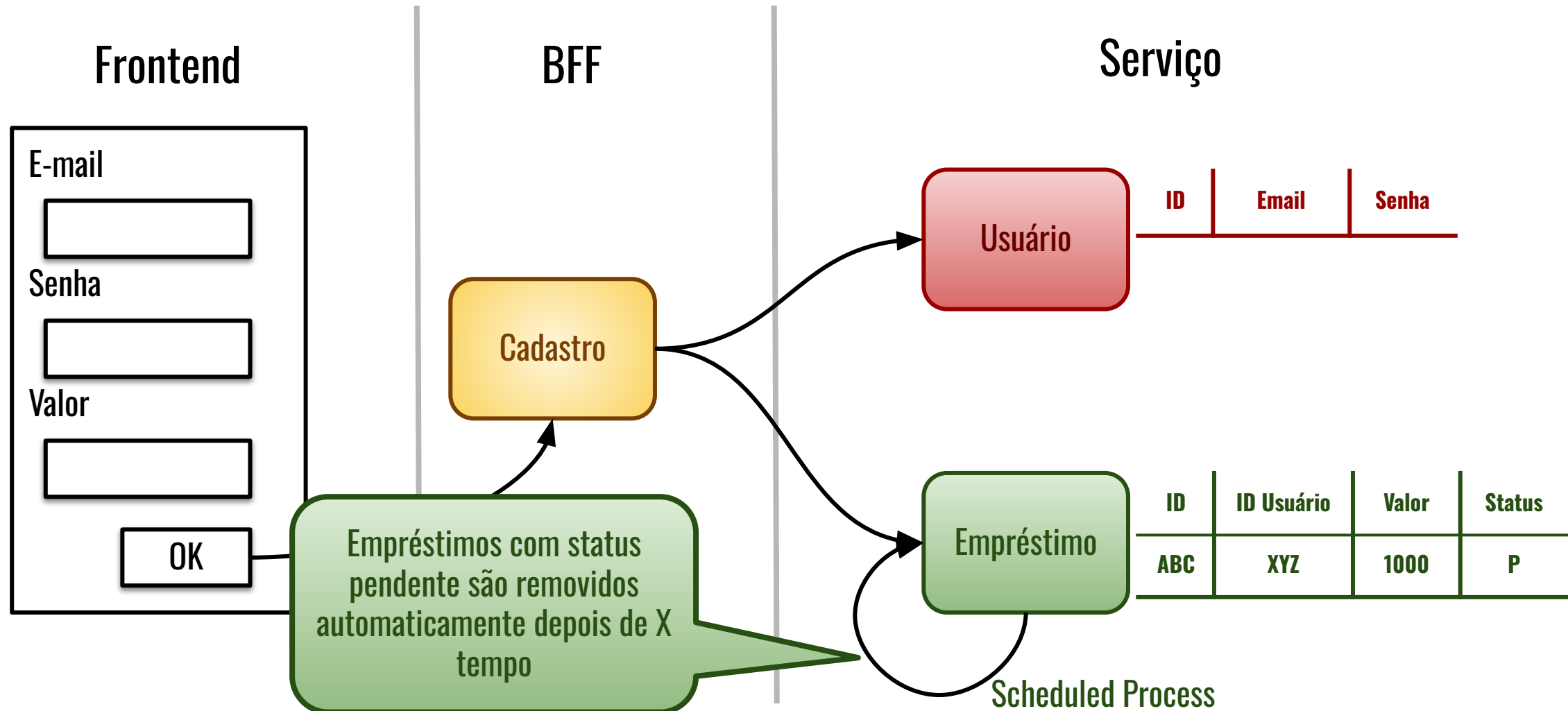
Modelando sem transação



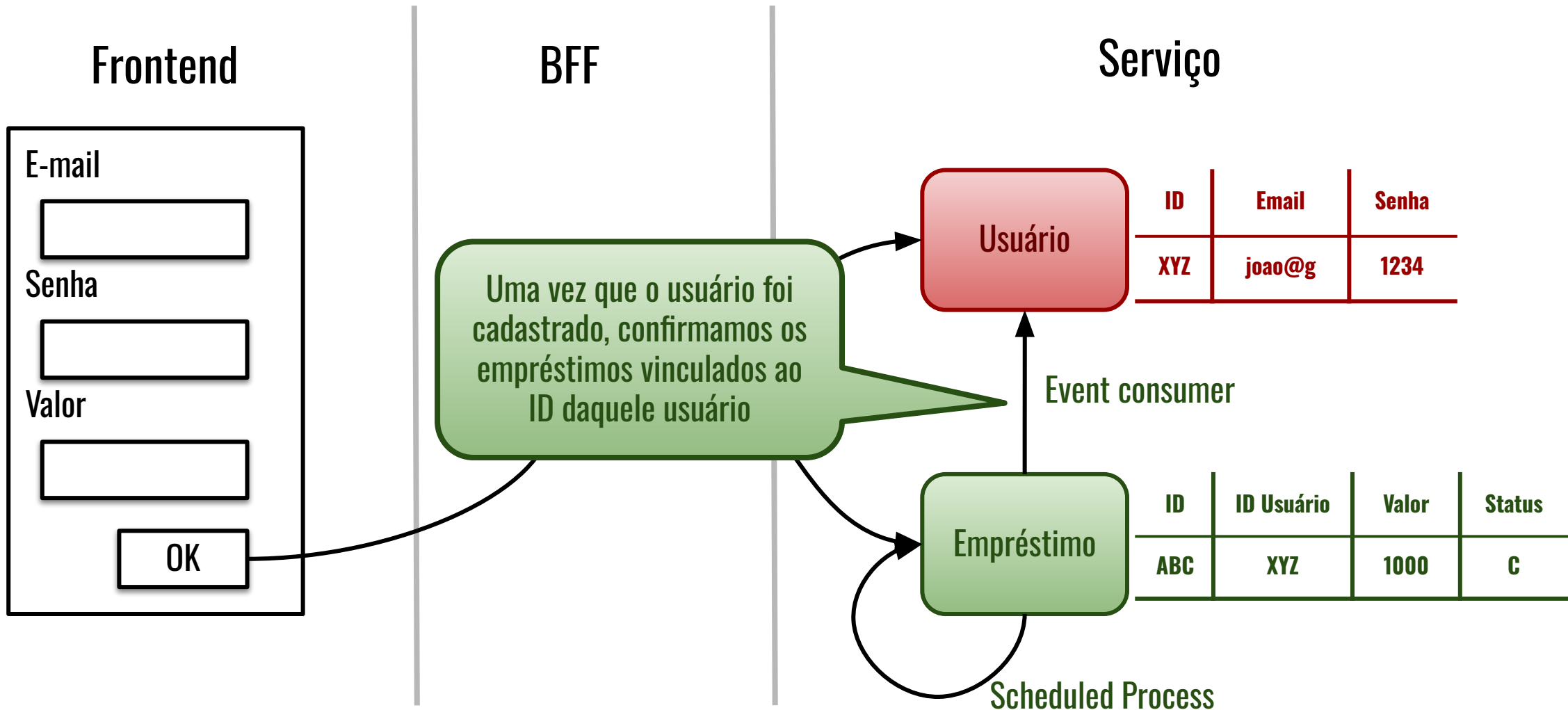
Modelando sem transação



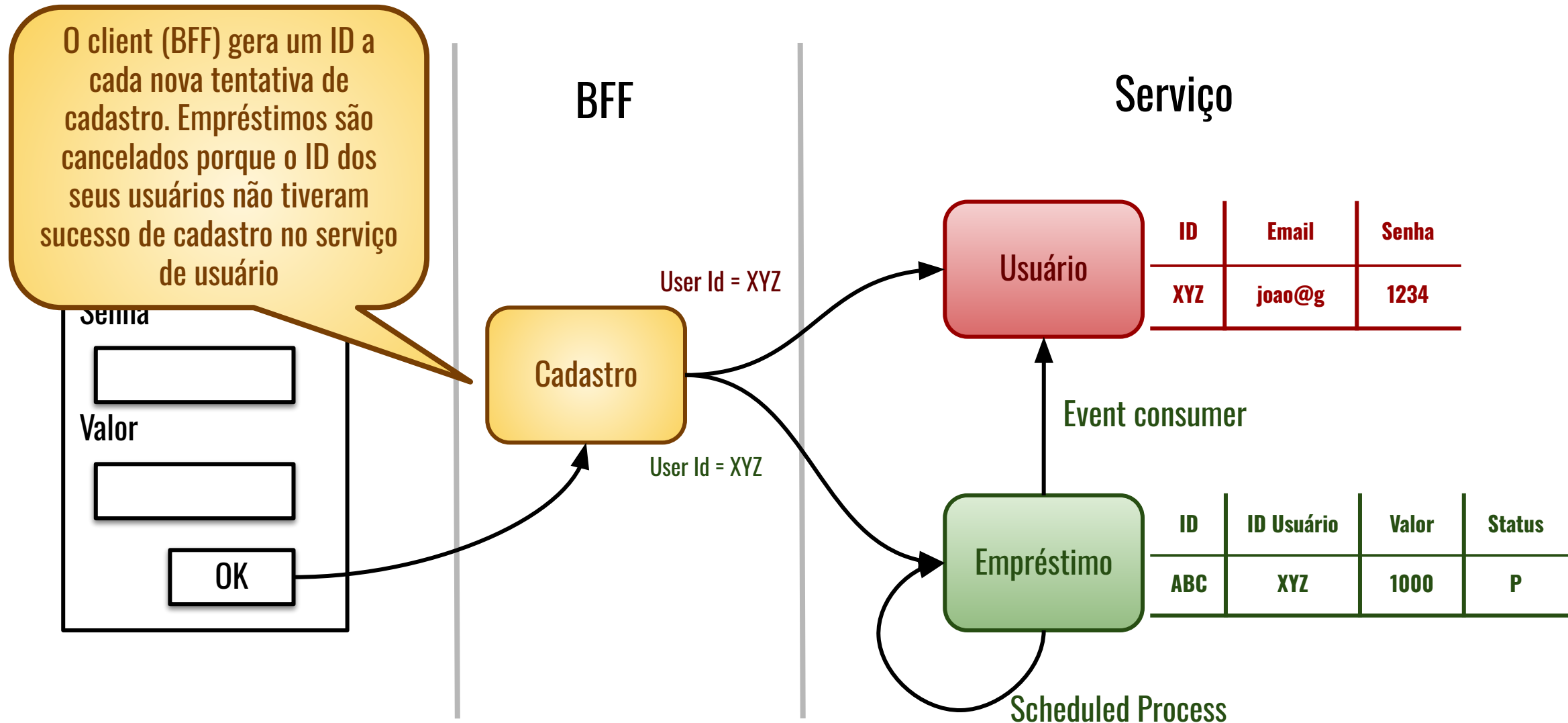
Modelando sem transação



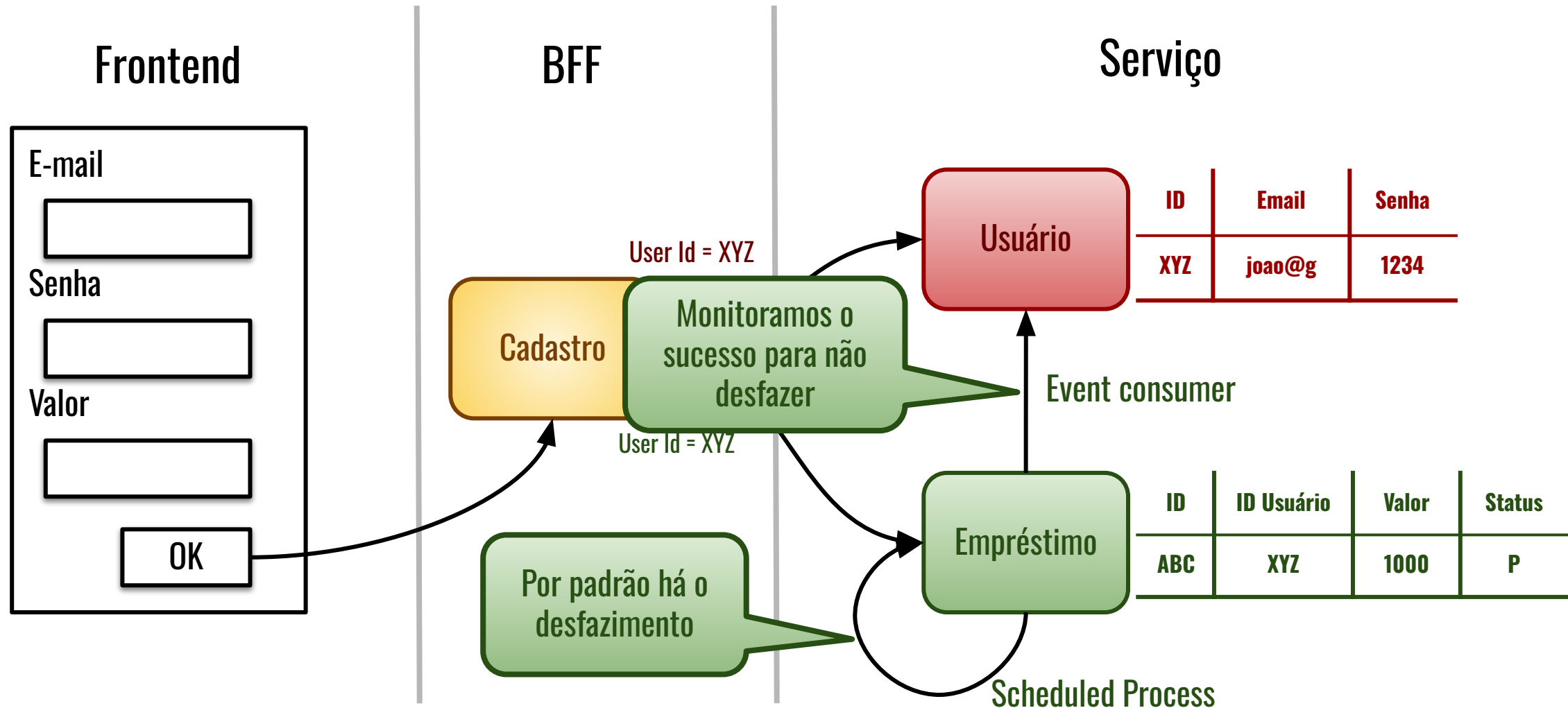
Modelando sem transação



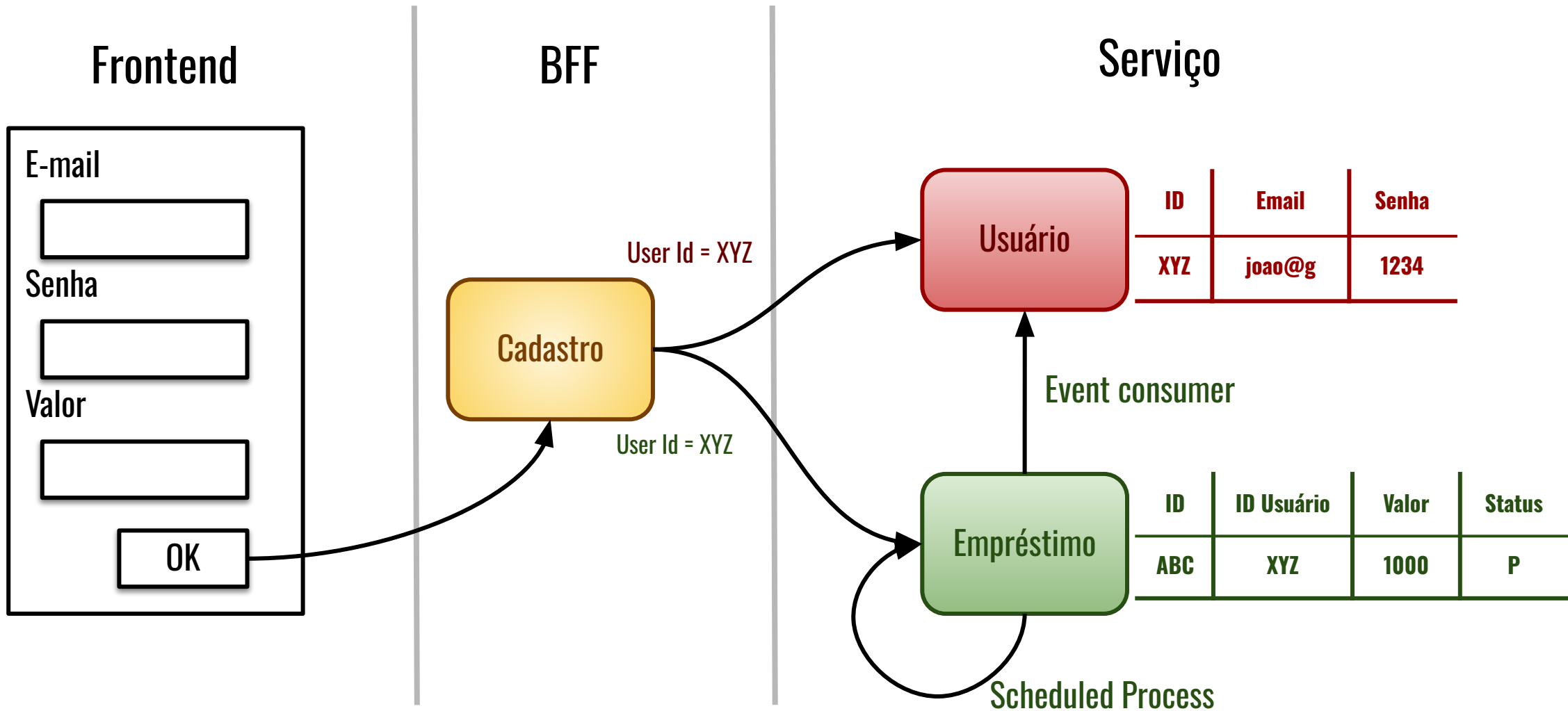
Modelando sem transação



Modelando sem transação

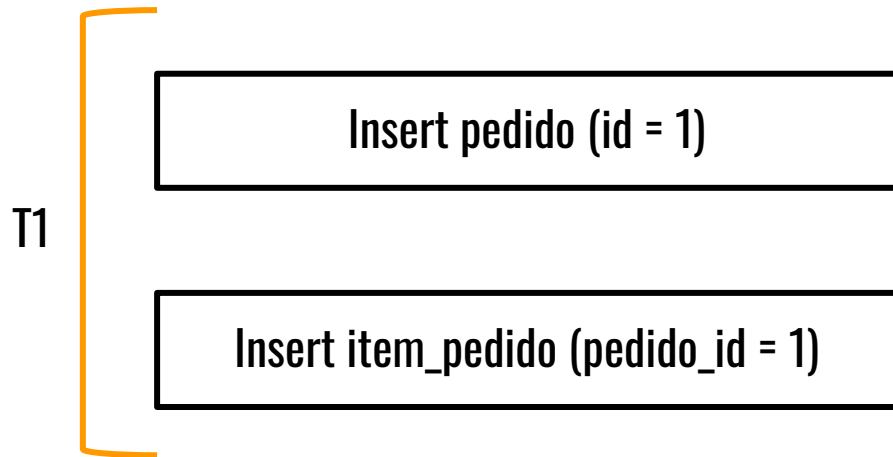


Modelando sem transação

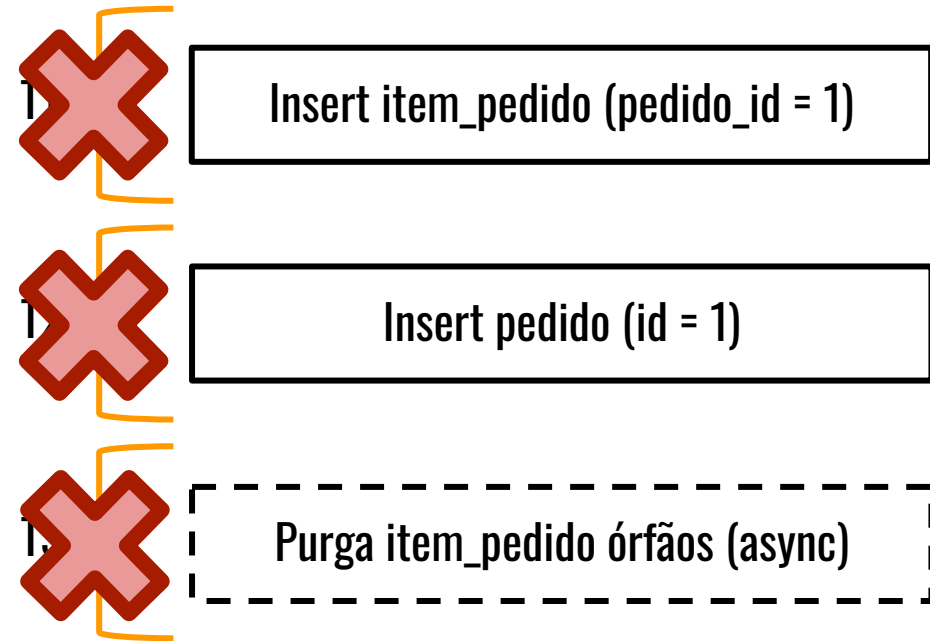


Modelando sem transação

Com transação

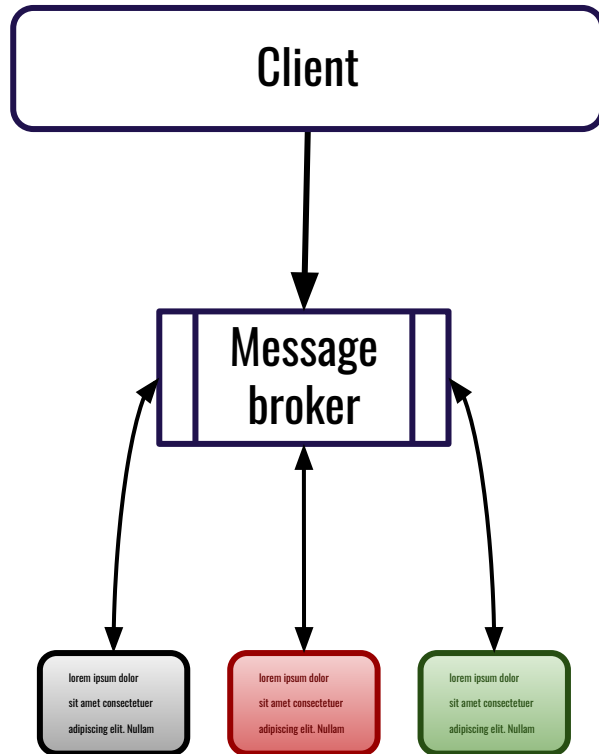


Sem transação

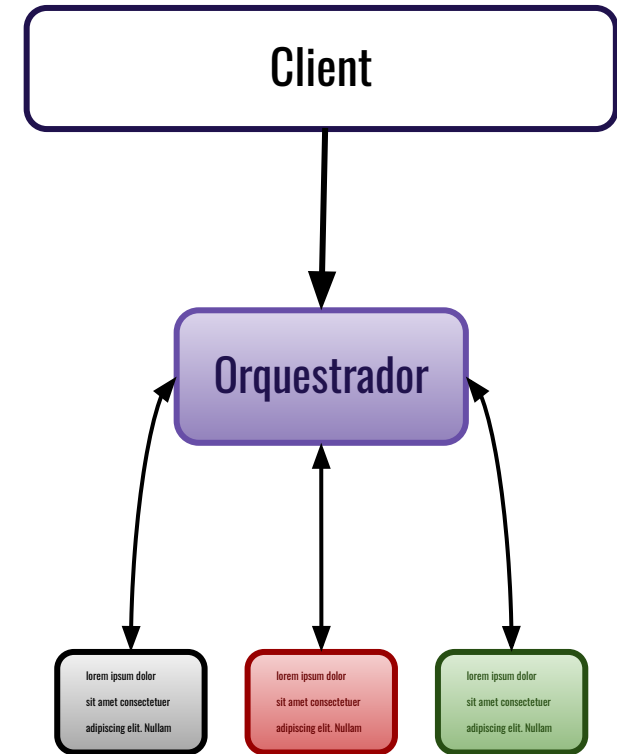


Saga Pattern

Coreografia



Orquestração

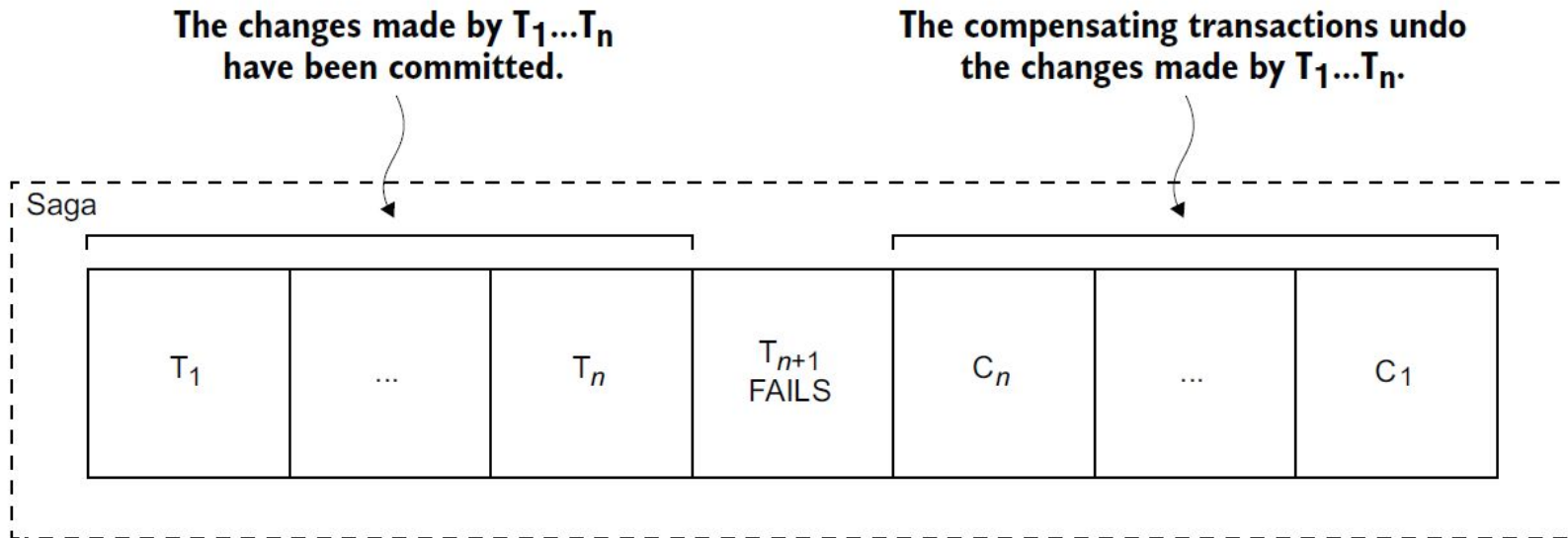


Sagas e Transações Compensadoras

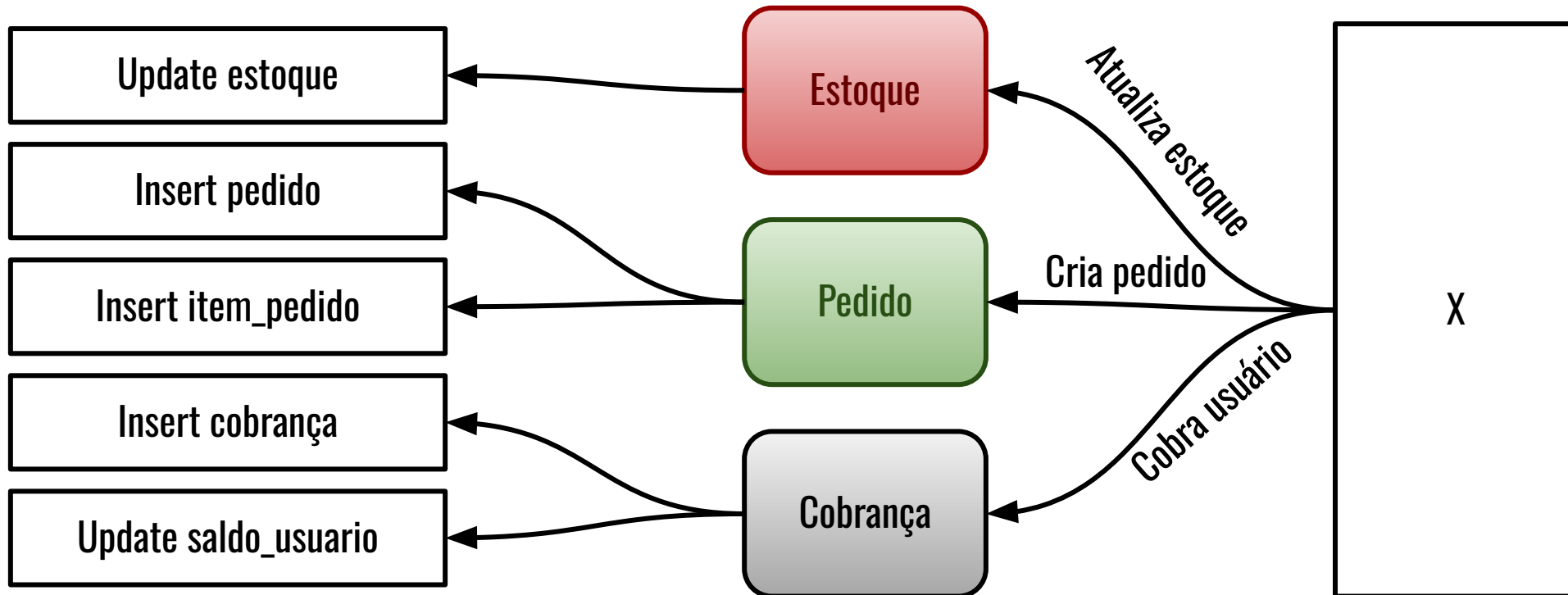
Um ótimo recurso das transações ACID é a facilidade para se fazer rollback caso haja uma violação na regra de negócios;

Infelizmente, Sagas não podem fazer roll back automaticamente, porque cada commit é feito em bancos locais;

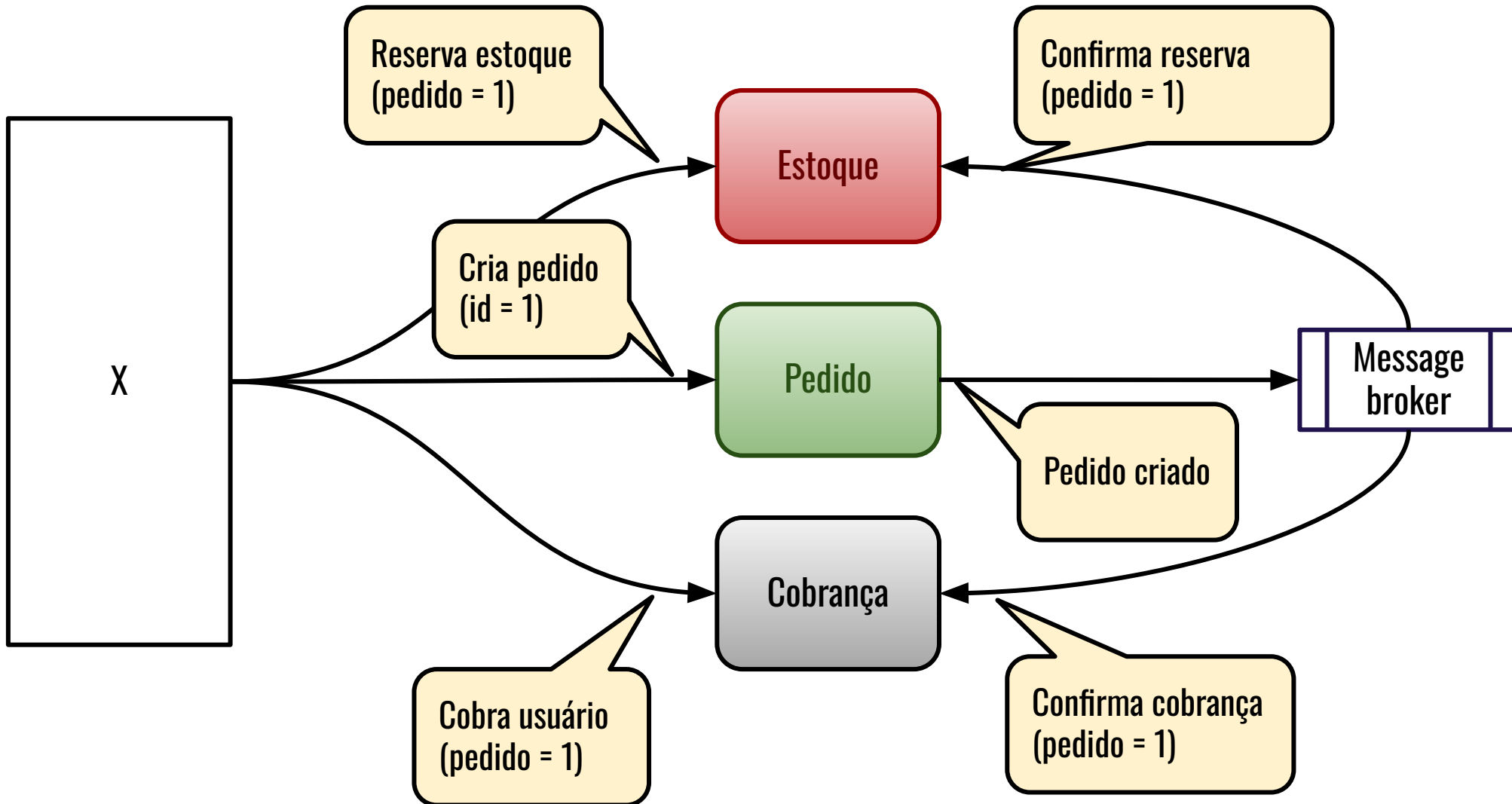
Conceitualmente, cada uma das etapas T_i , tem uma transação compensadora correspondente C_i que desfaz os efeitos da T_i .



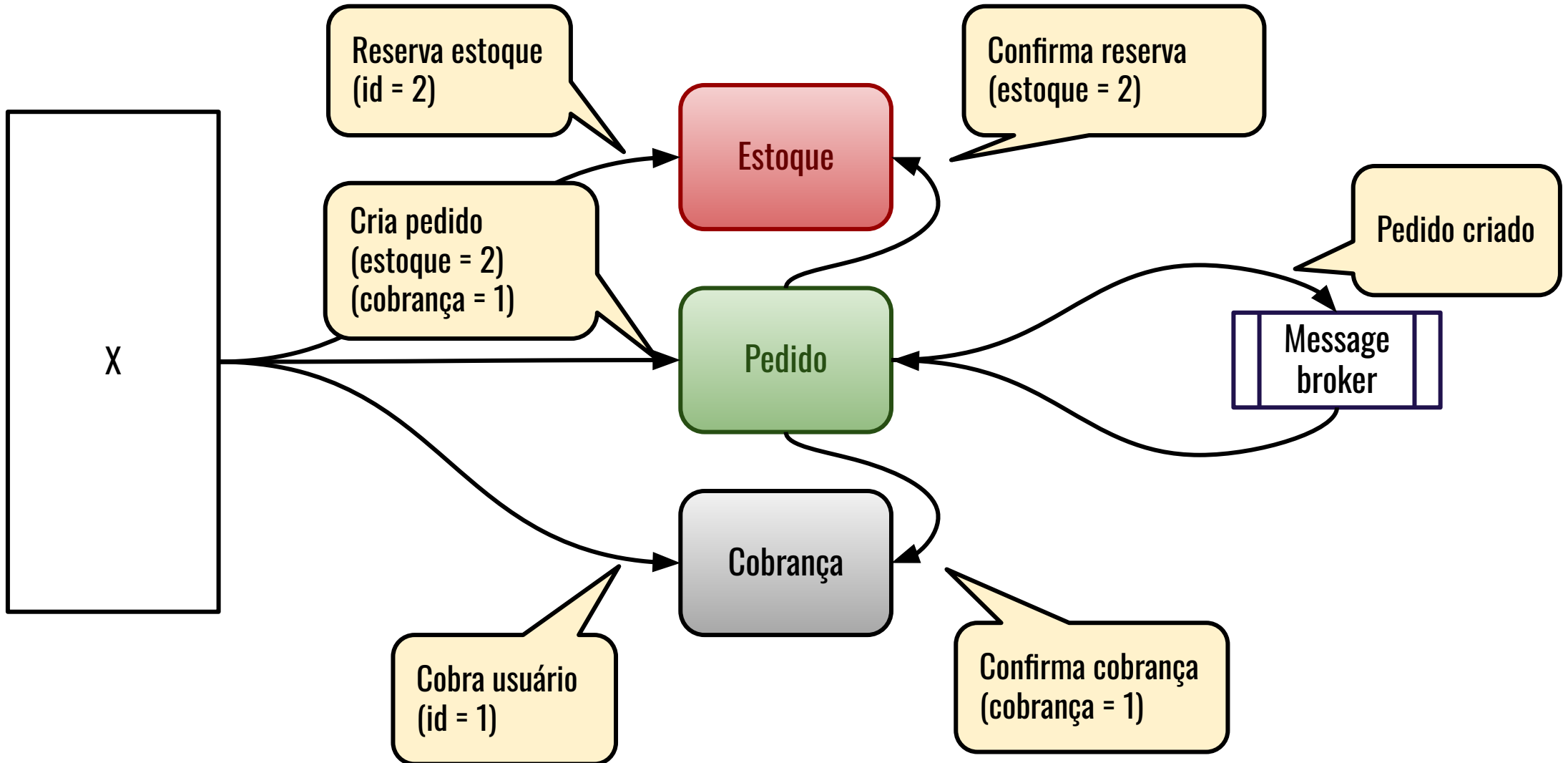
Saga Pattern



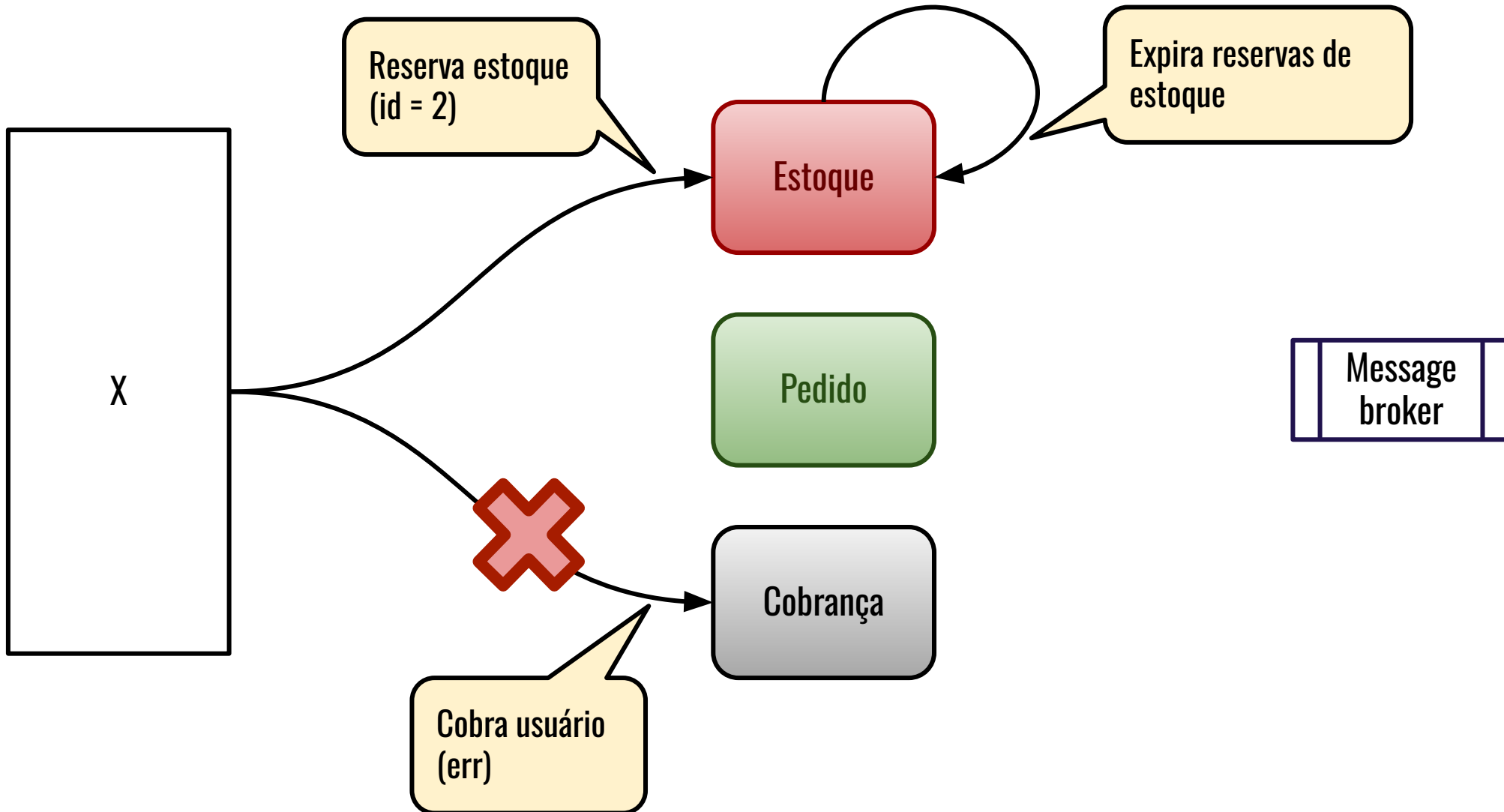
Saga Pattern



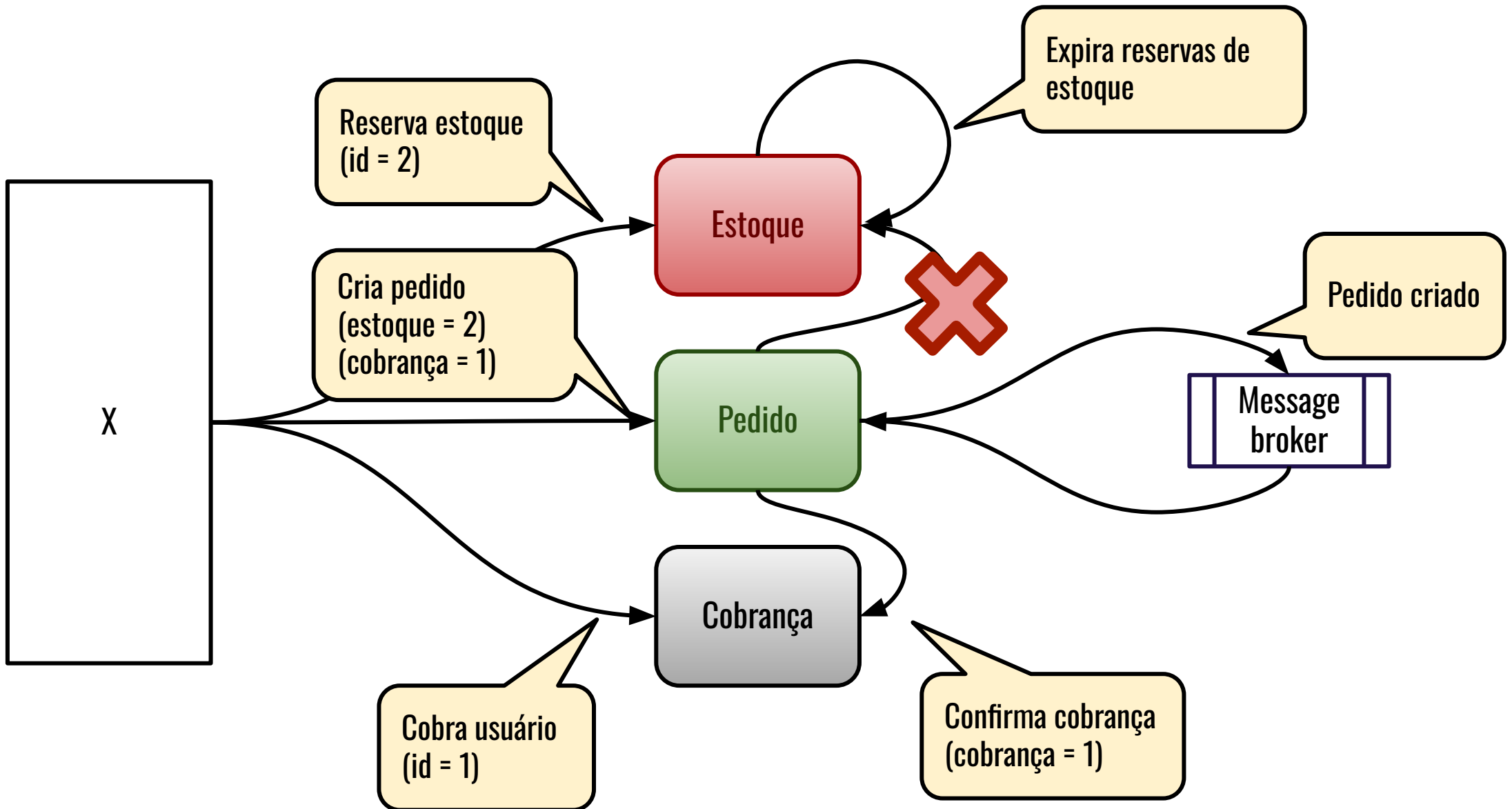
Saga Pattern



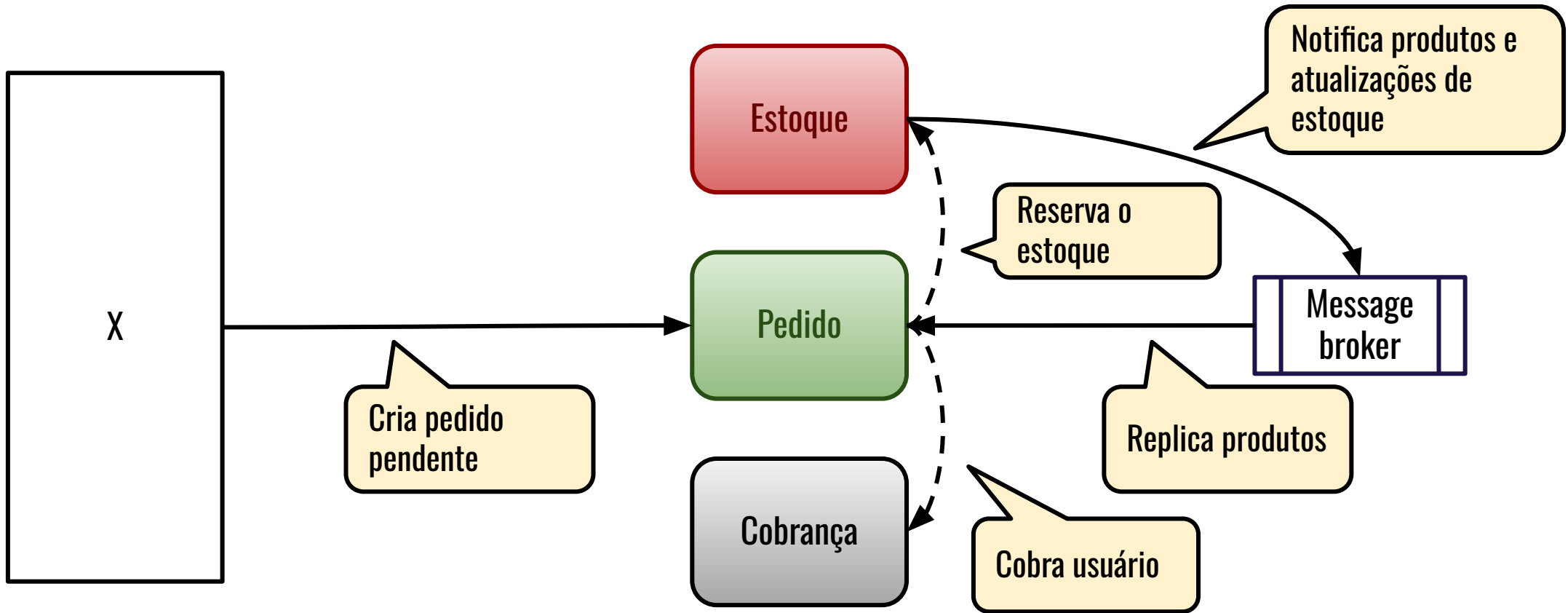
Saga Pattern



Saga Pattern



Saga Pattern



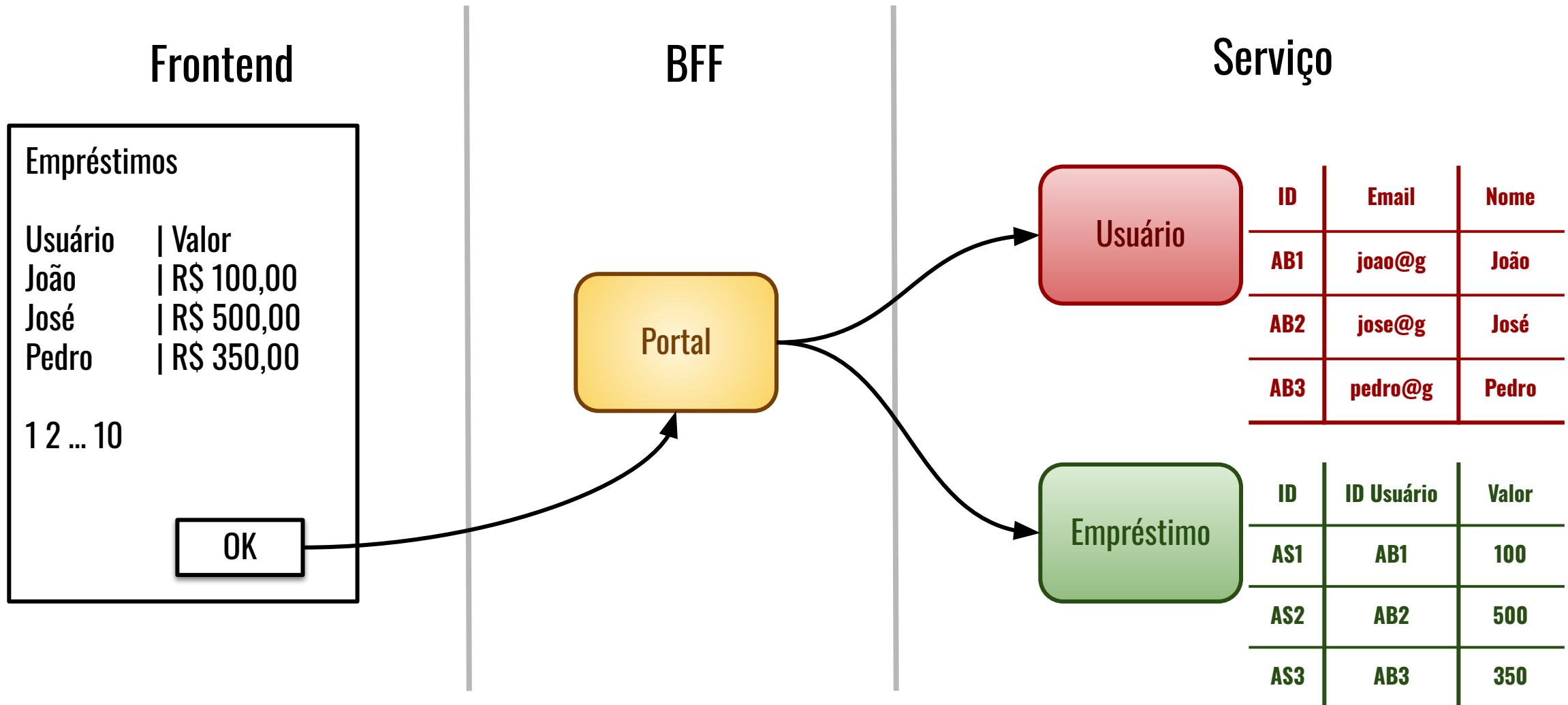
Algumas implementações

Netflix Conductor <https://netflix.github.io/conductor>

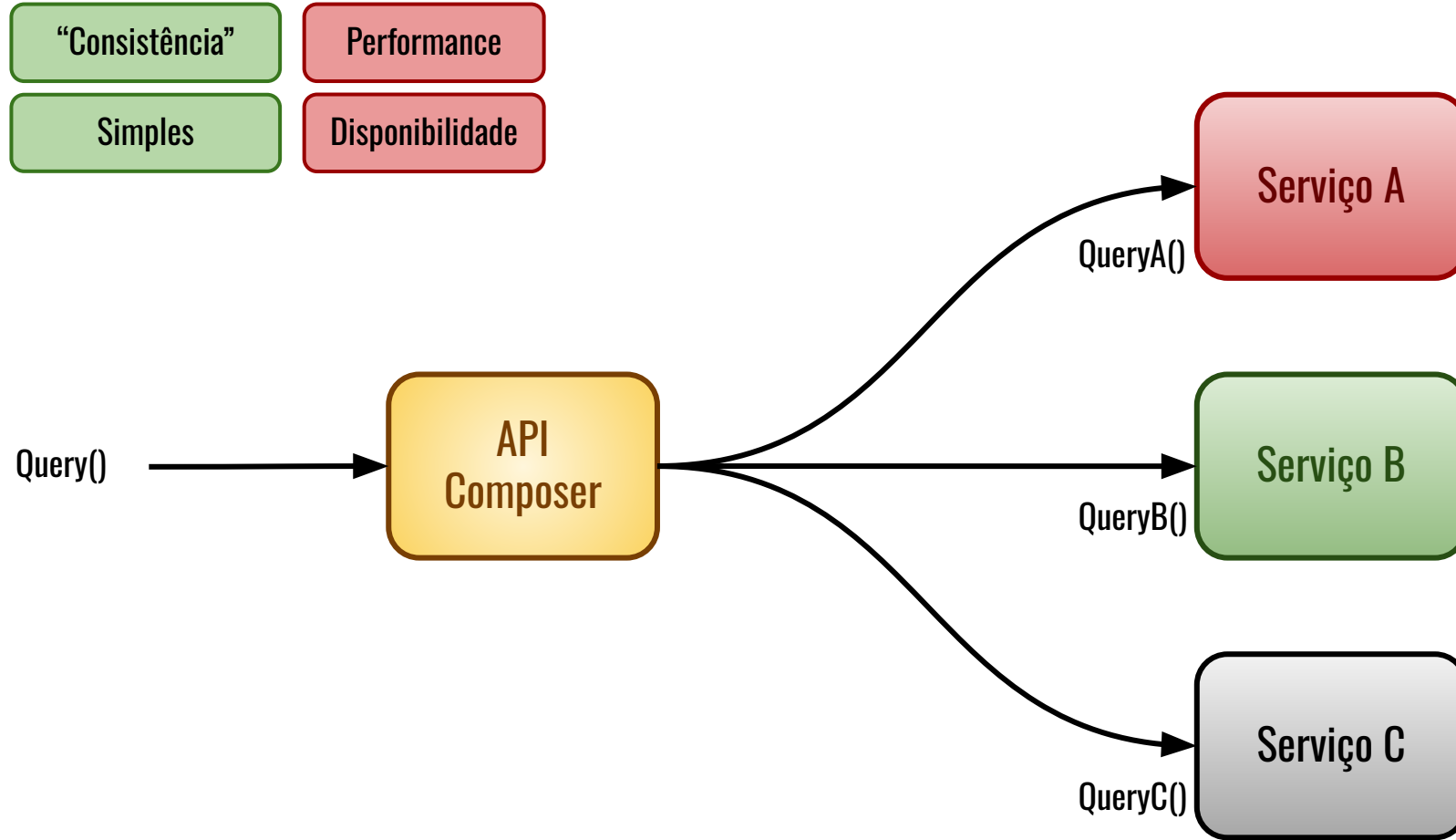
Spring Cloud Data Flow <https://spring.io/projects/spring-cloud-dataflow>

Zeebe <https://zeebe.io>

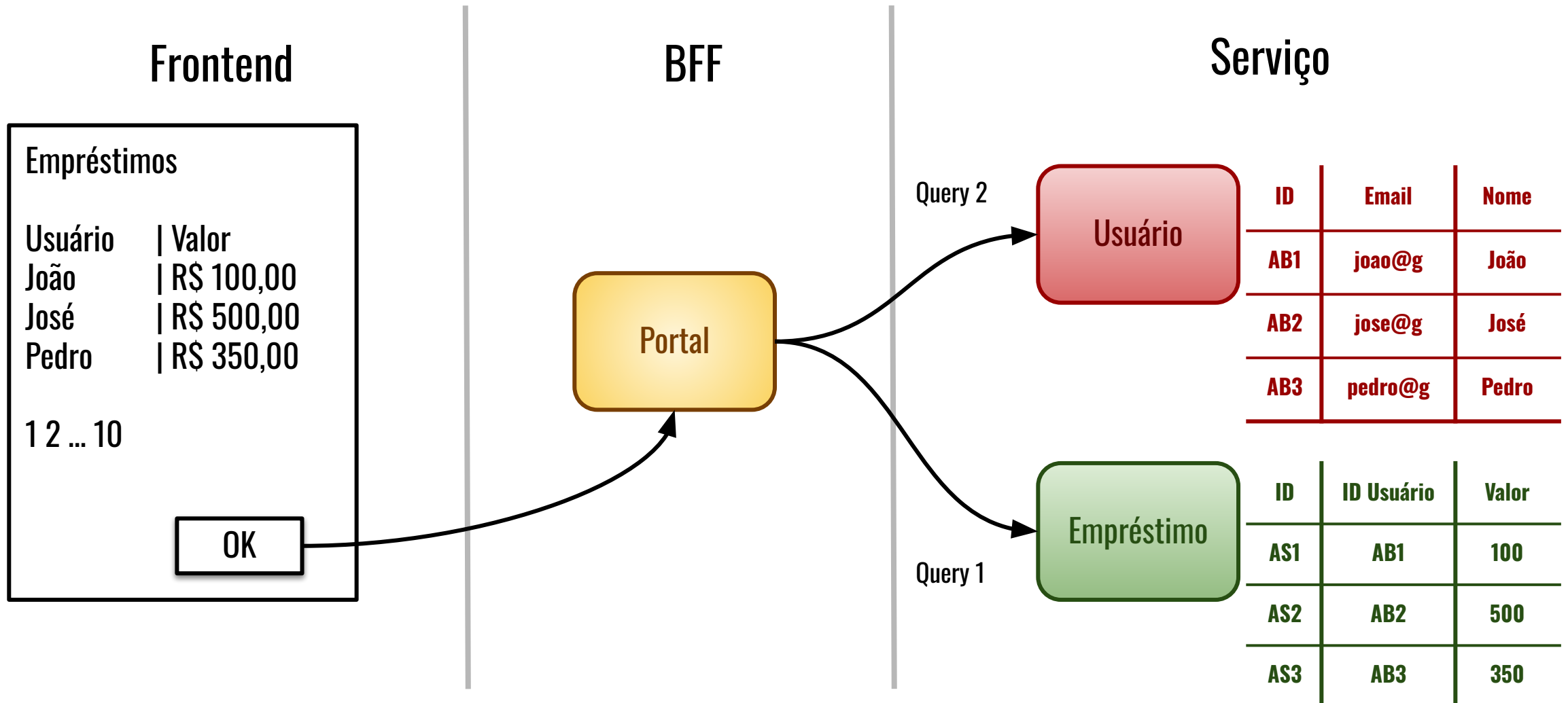
Queries nos microserviços



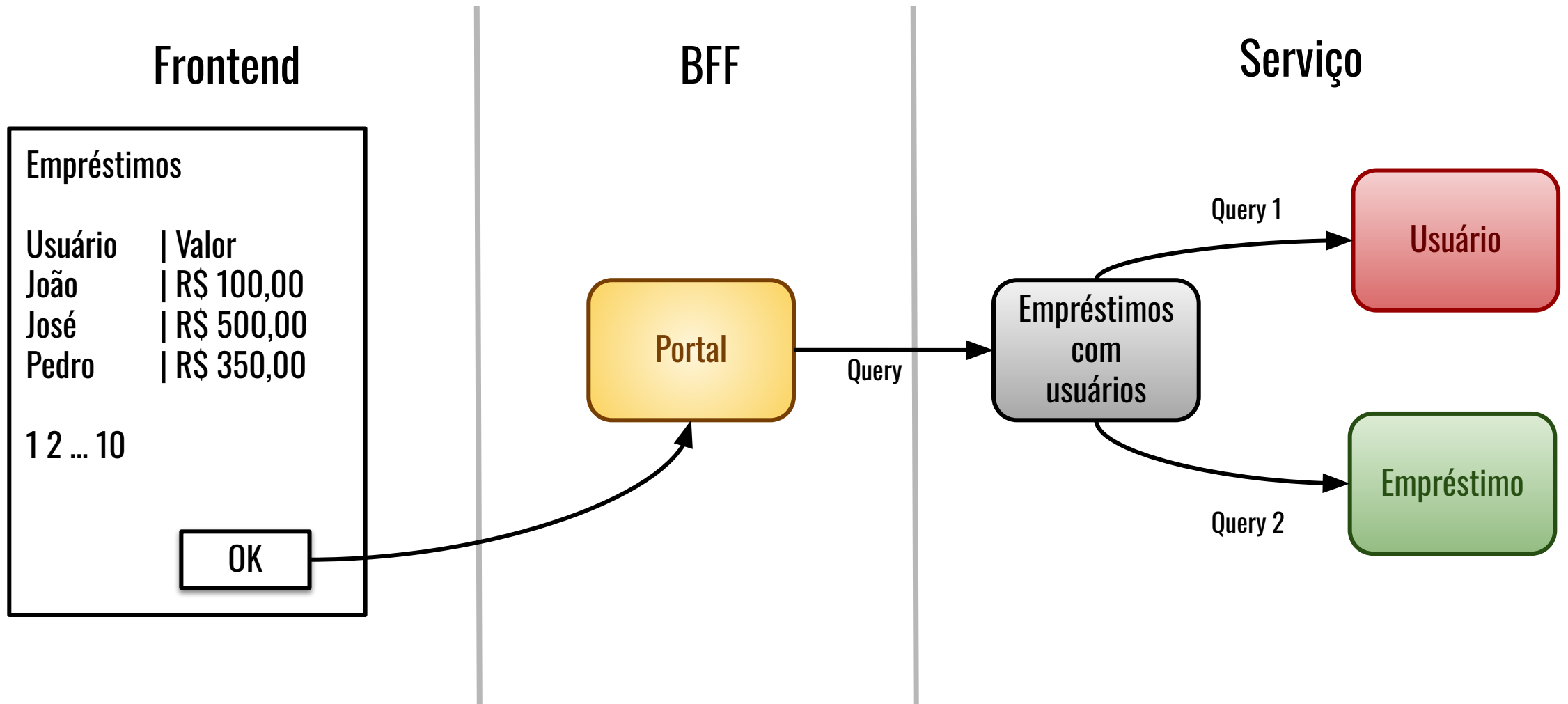
Composition Pattern



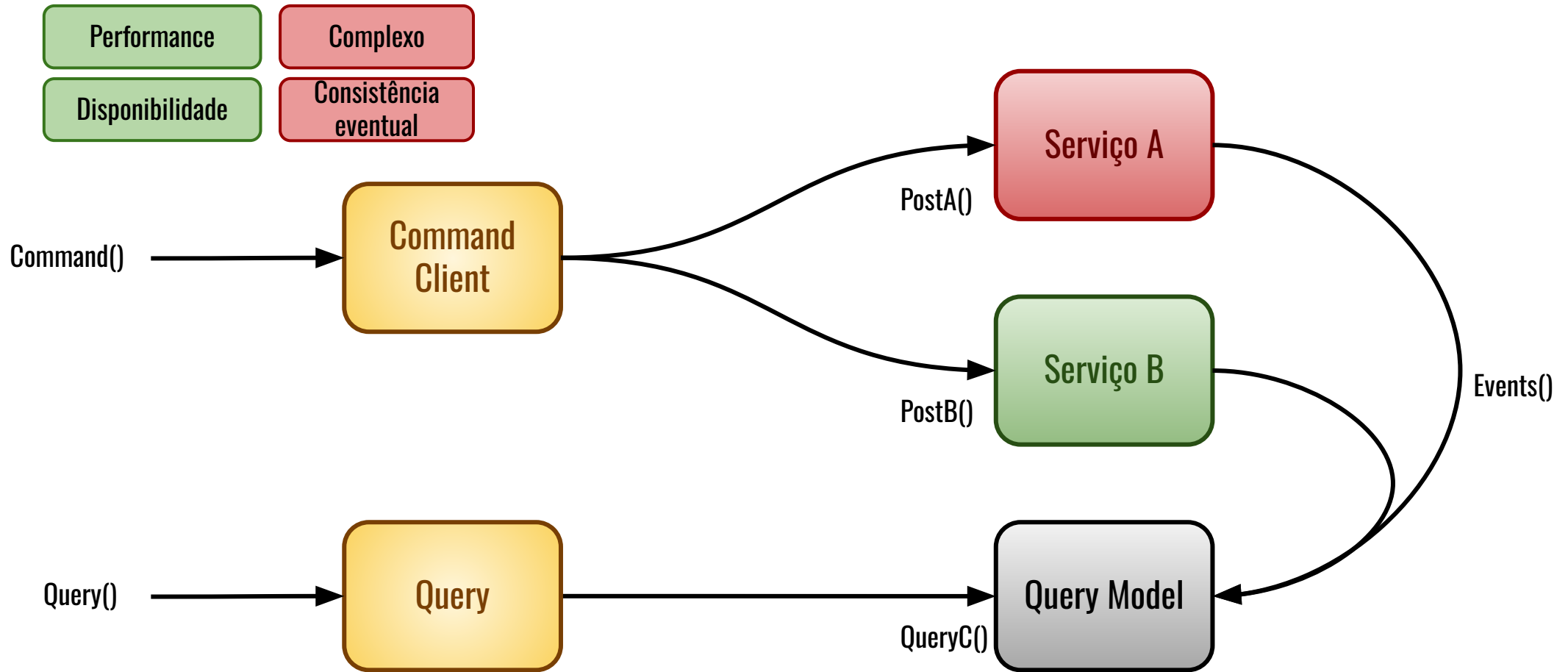
Composition Pattern



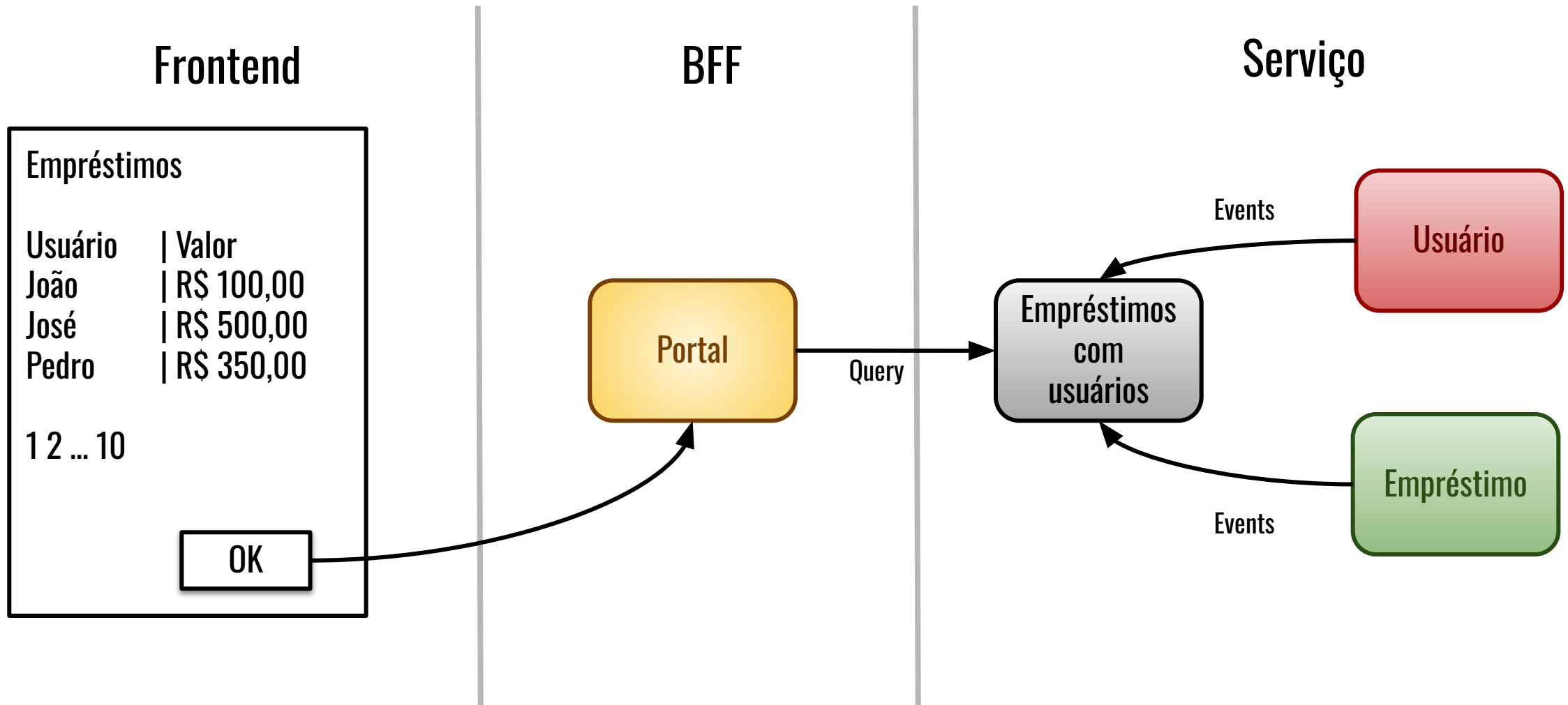
Composition Pattern



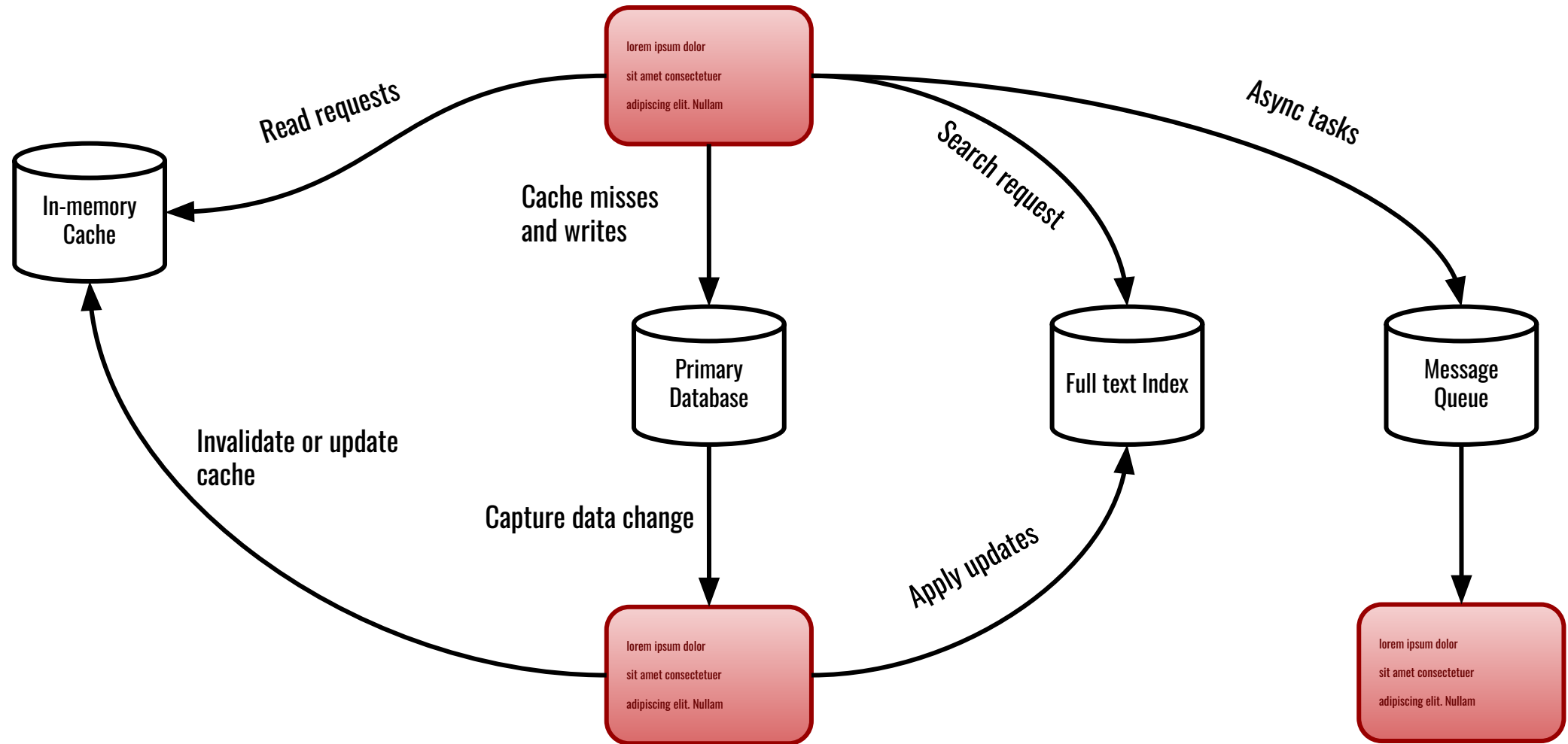
Command Query Responsibility Segregation



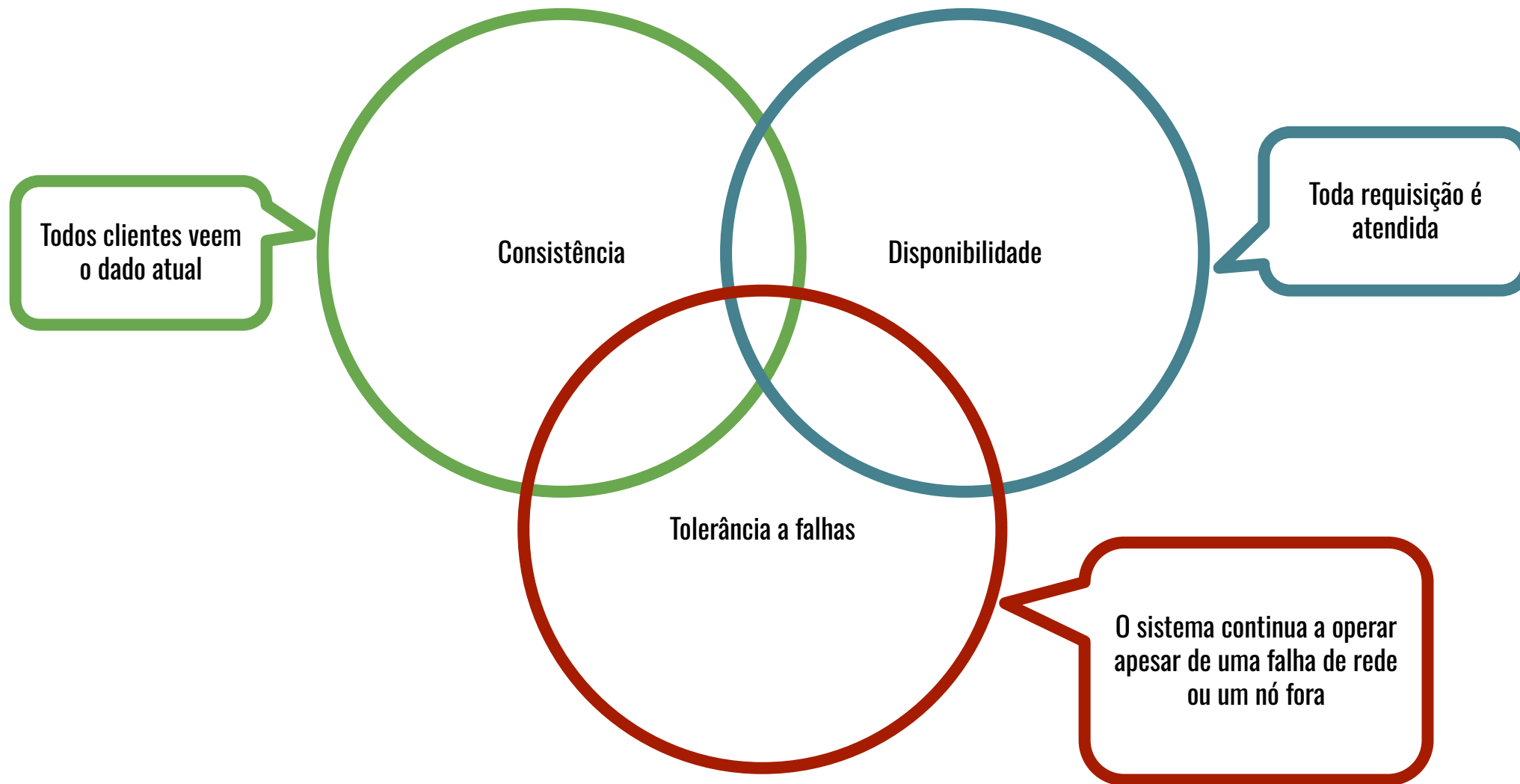
Command Query Responsibility Segregation



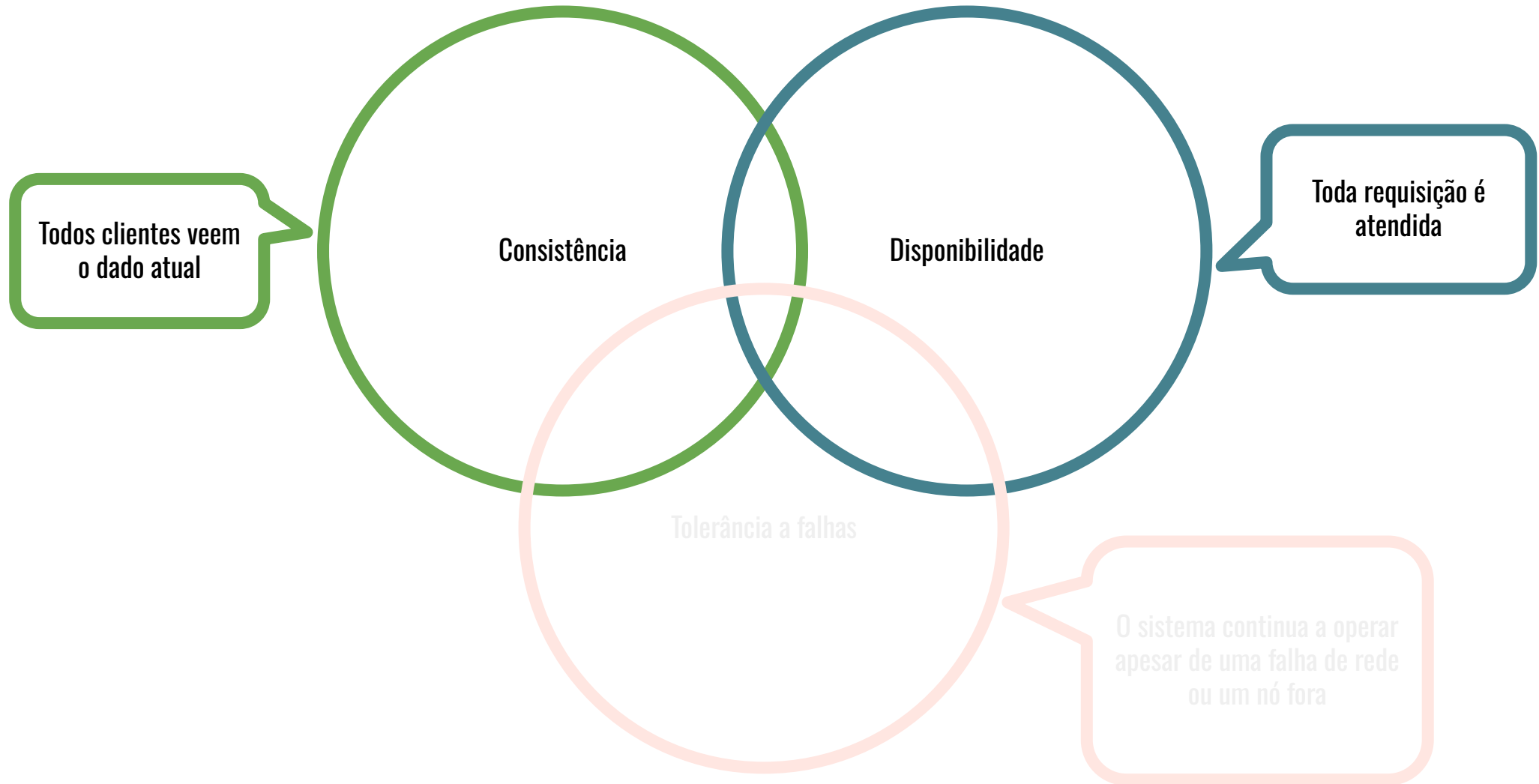
Sistemas de dados



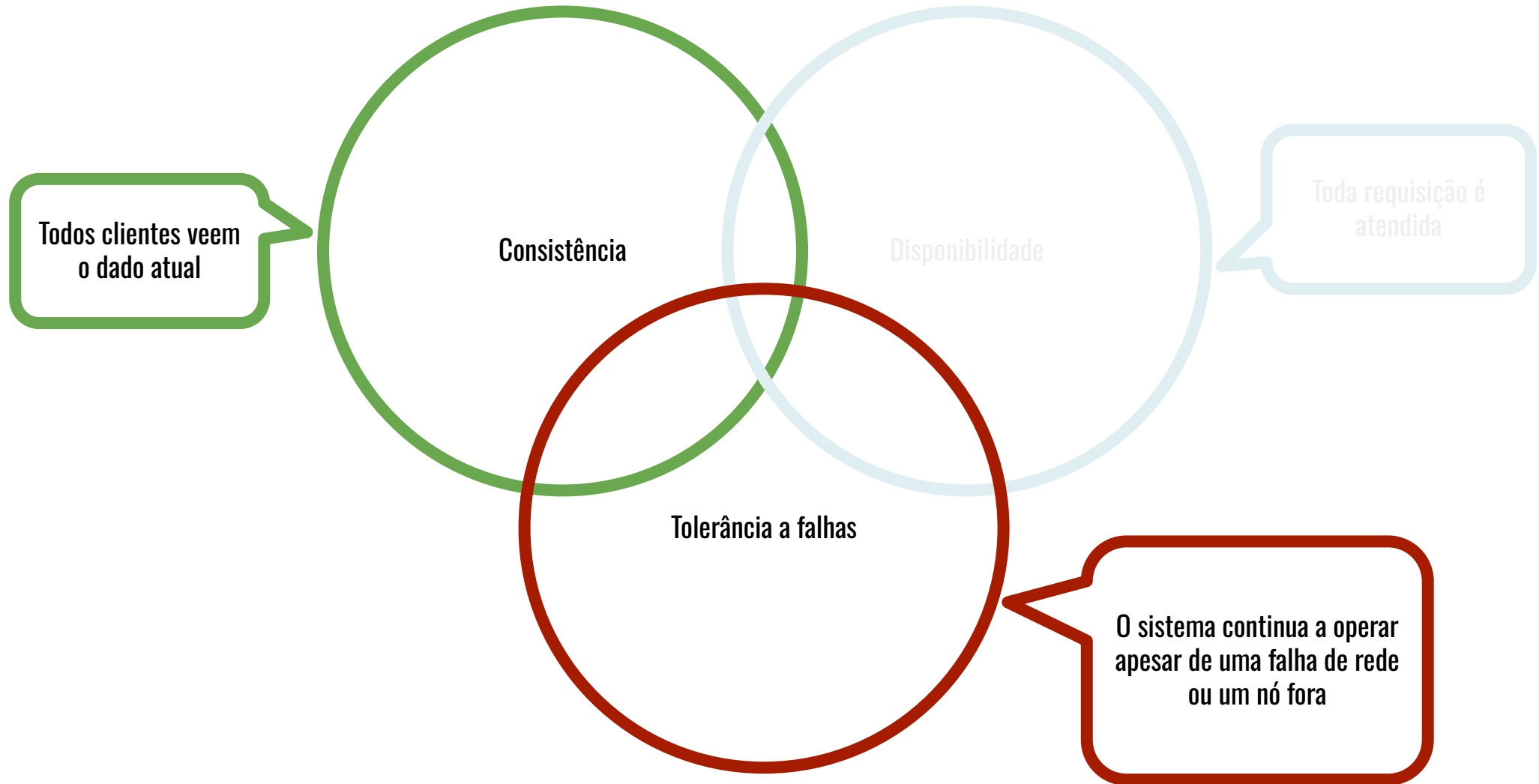
Teorema CAP



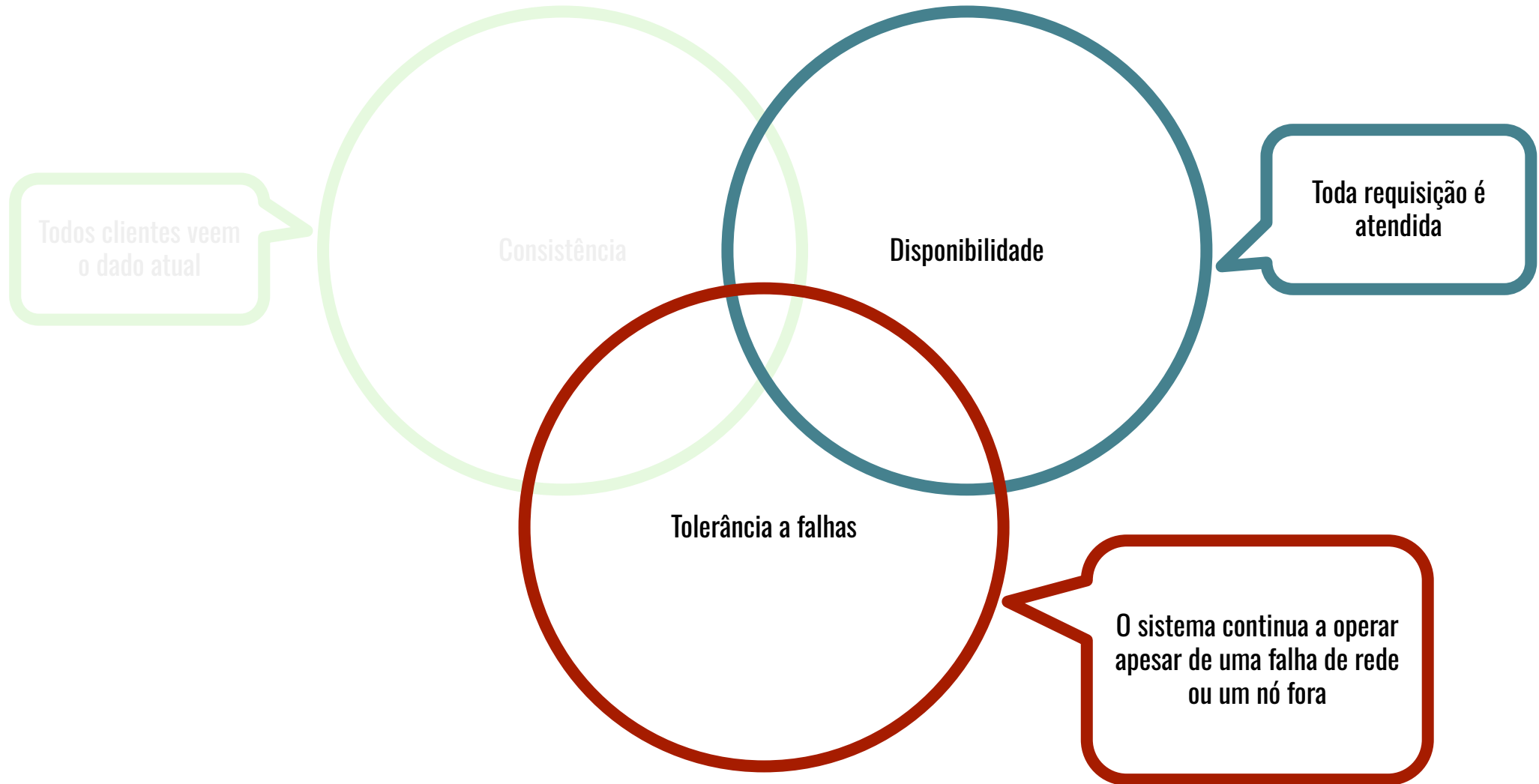
Teorema CAP (CA)



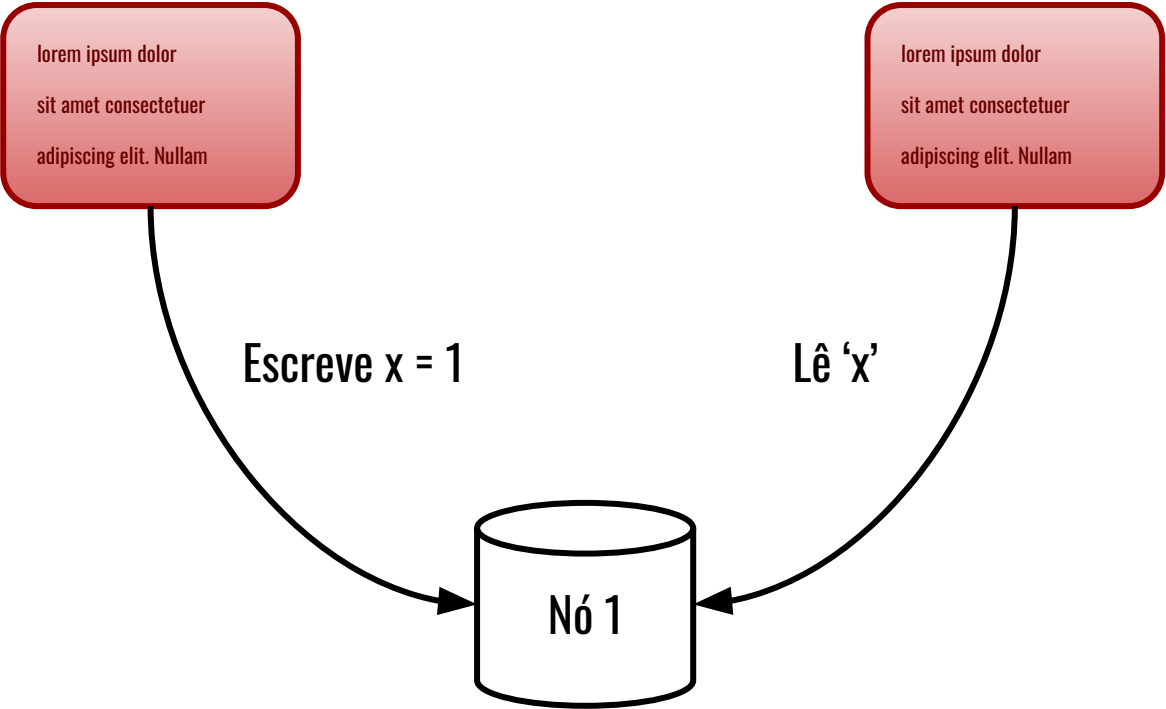
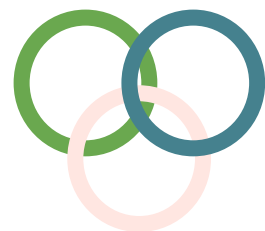
Teorema CAP (CP)



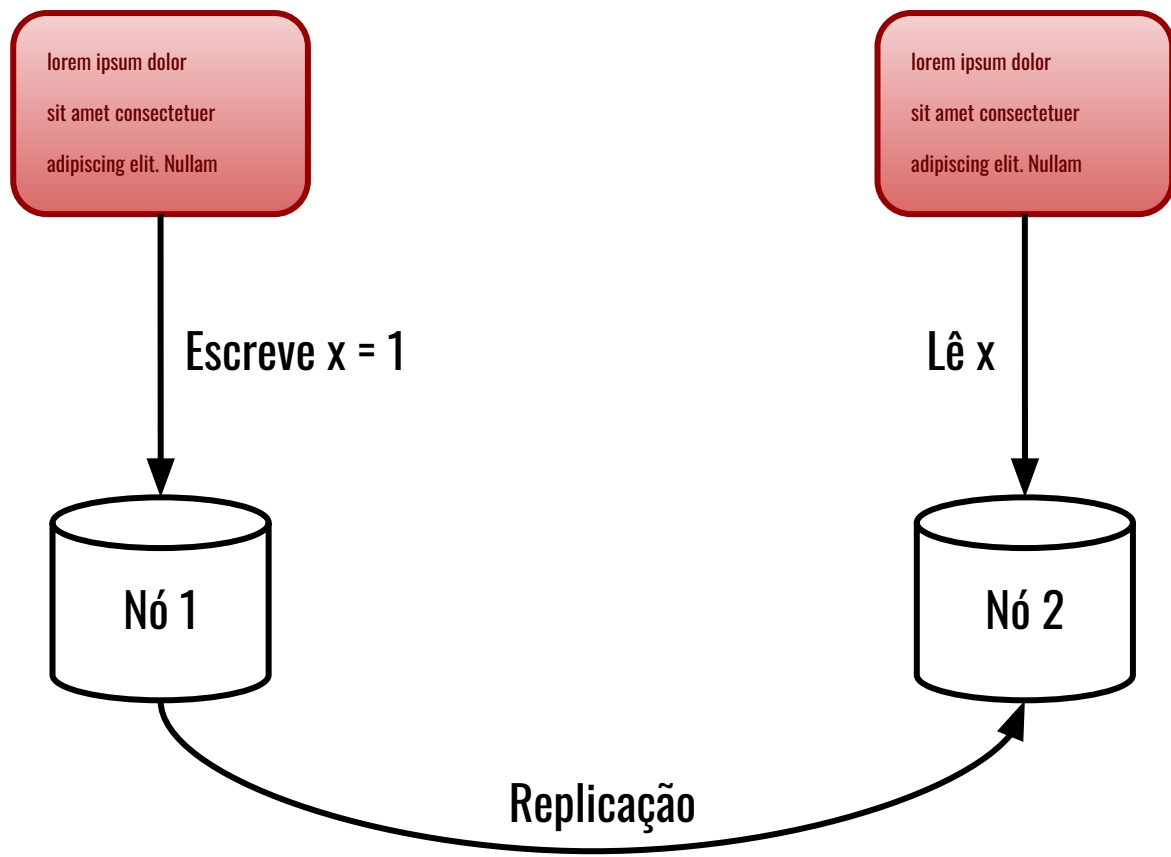
Teorema CAP (AP)



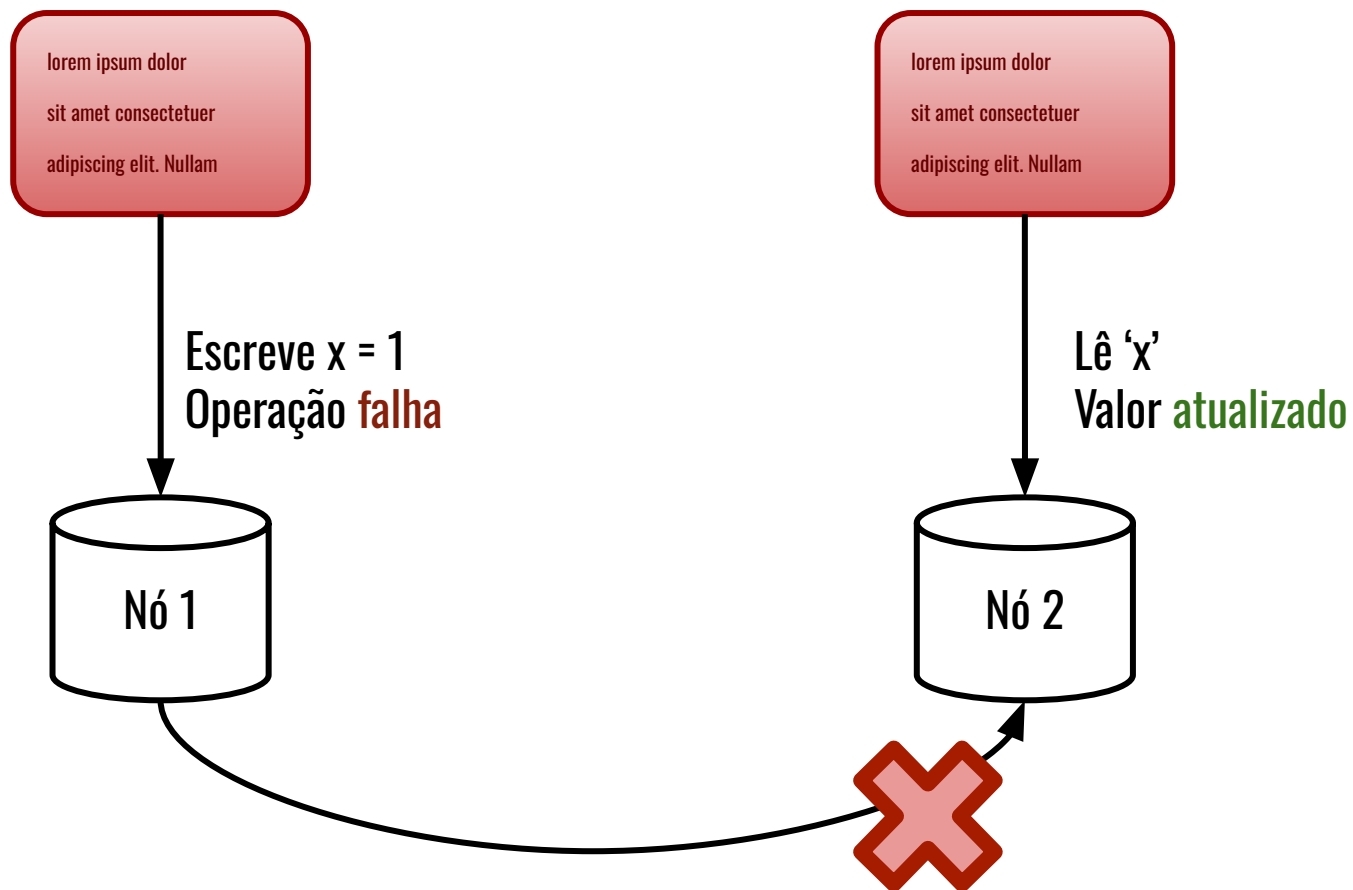
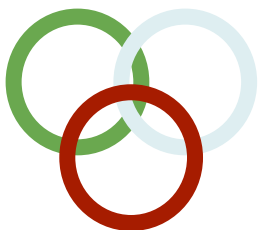
Teorema CAP



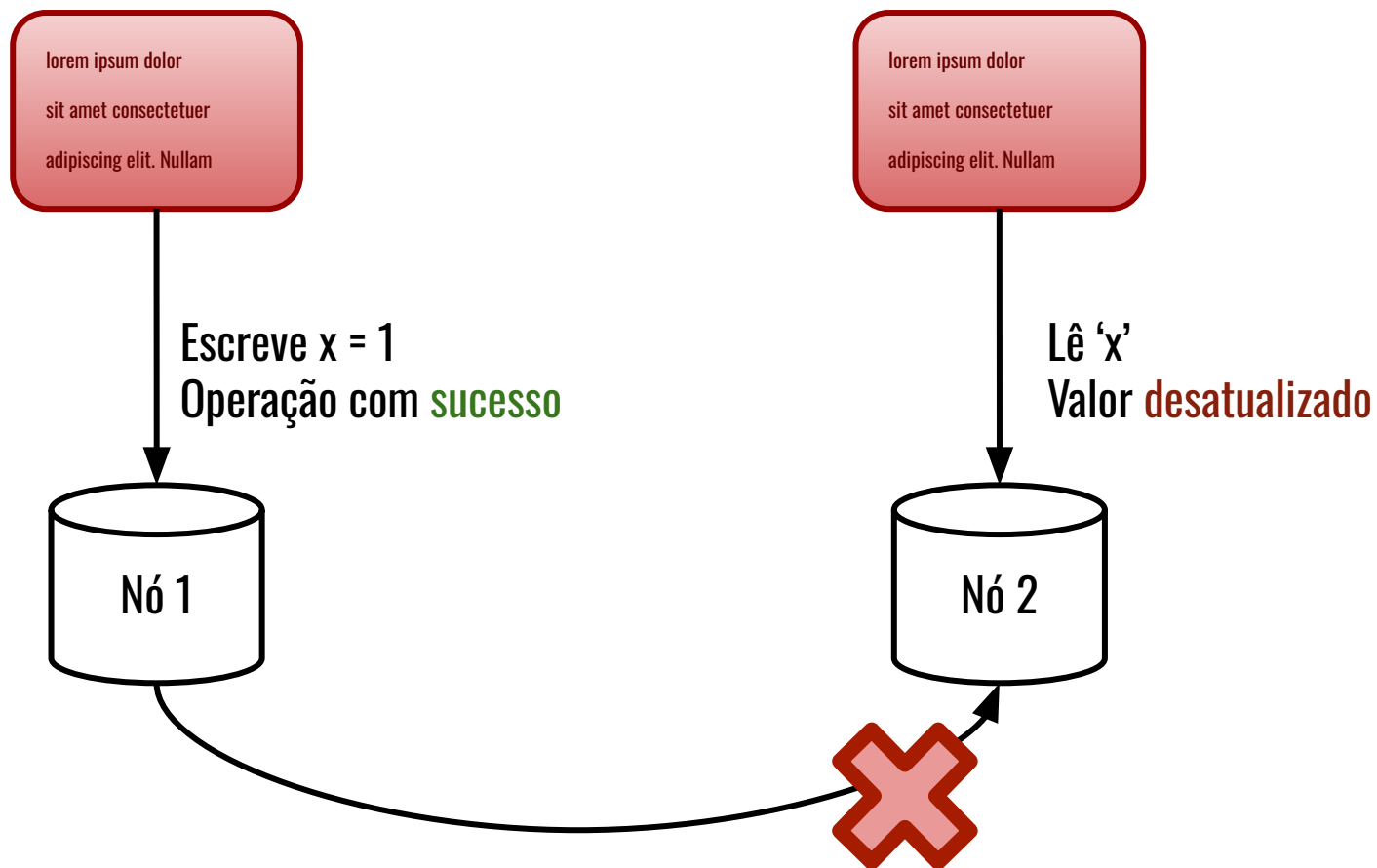
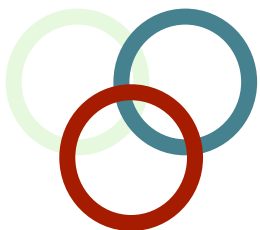
Teorema CAP



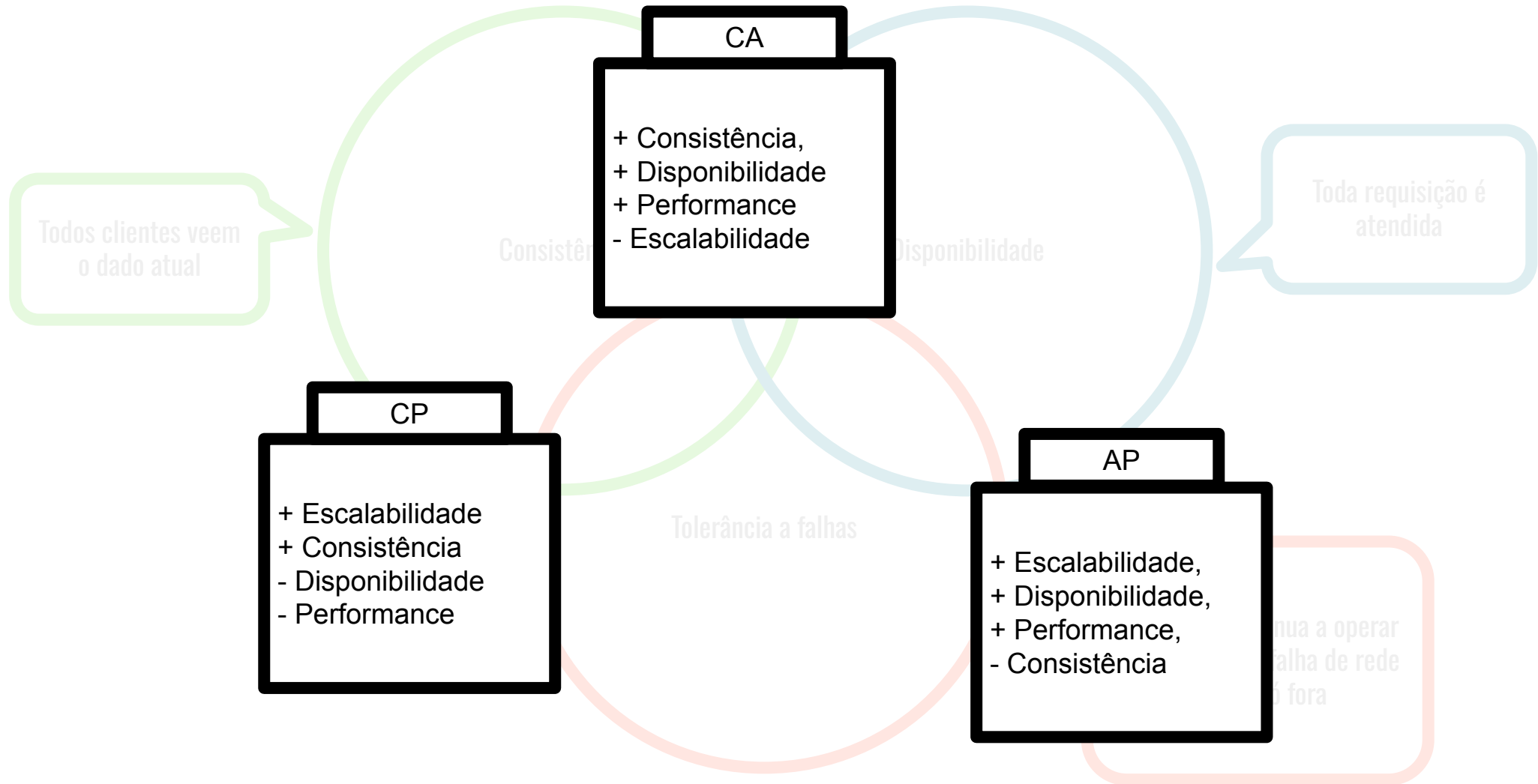
Teorema CAP



Teorema CAP



Teorema CAP



Dúvidas